ROAD DESIGN - PRELIMINARY DESIGN DRAWINGS

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Standards profile: US - AASHTO Green Book 2018 / FHWA Design speed 50 km/h

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GENERAL NOTES

1. Design to US - AASHTO Green Book 2018 / FHWA - parameter register lists each value as verified (from the published standard), formula, or engineer input.
2. Values from copyrighted or paywalled tables are not shipped with the software; they are entered by the engineer and marked as engineer inputs in the parameter register and design report.
3. Dimensions in metres unless noted; elevations in metres on the survey datum; stations in km+m along the control line.
4. Existing ground from the supplied survey/terrain points (TIN). Verify on site before construction.
5. Earthwork volumes by average end area with a composite-grid cross check; shrink/bulk factors and haul limits as stated in the earthwork report.
6. Standard details are schematic references; construct to the cited agency standard drawings and specification.
7. These drawings are generated by agent tools from the audit-logged design. Status: PRELIMINARY unless the engineer has recorded approval; sealing is the engineer's act.
8. Selected route: A-direct (recorded with reasons in the audit log).

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cover, sheet index and general notes

Sheet 1 of 32

Scale: NTS (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

175bc80d01efaeeb29d3307375fe64e2

a486d4e6670d7de3a340c8953107d8d8

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

PRELIMINARY

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Alternative	Length m	Min R m	Max grade %	Cut m3	Fill m3	Imbalance m3	Wall conc. m3	Mandatory fails	Cost proxy
A-direct	1136	180	8.00	22618	51310	-30954	6581	0	317776
B-contour-following	1088	85	8.00	39482	69081	-33548	4691	3	299615
C-separate-corridor	1129	185	8.00	20967	100404	-81533	77824	0	2578395

Selection rule: fewest mandatory geometry failures, then lowest cost proxy (unit rates are agent placeholders unless the engineer enters rates)
- vs B-contour-following: cost proxy 317776 vs 299615 (+6.1%), length 1136 vs 1088 m, cut+fill 73928 vs 108563 m³, imbalance -30954 vs -33548 m³, wall concrete 6581 vs 4691 m³, max grade 8.0 vs 8.0 %, mandatory failures 0 vs 3
- vs C-separate-corridor: cost proxy 317776 vs 2578395 (-87.7%), length 1136 vs 1129 m, cut+fill 73928 vs 121371 m³, imbalance -30954 vs -81533 m³, wall concrete 6581 vs 77824 m³, max grade 8.0 vs 8.0 %, mandatory failures 0 vs 0
Cost proxy uses placeholder unit rates unless the engineer enters rates; it ranks alternatives and is not an estimate. Least-cost search on a 10 m grid, 16 headings, grade penalty above the profile maximum.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Route alternatives and selection
Sheet 2 of 32
Scale: 1:2500 (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
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Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

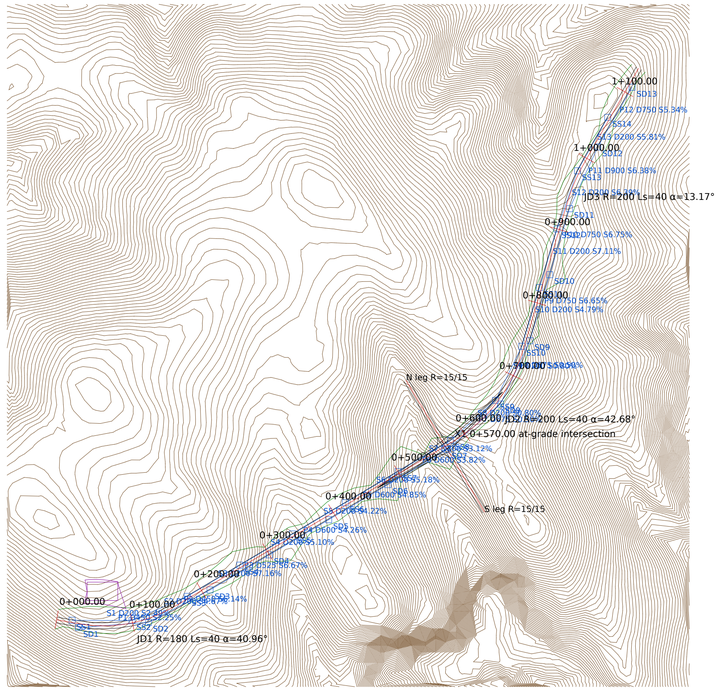
NOT FOR CONSTRUCTION

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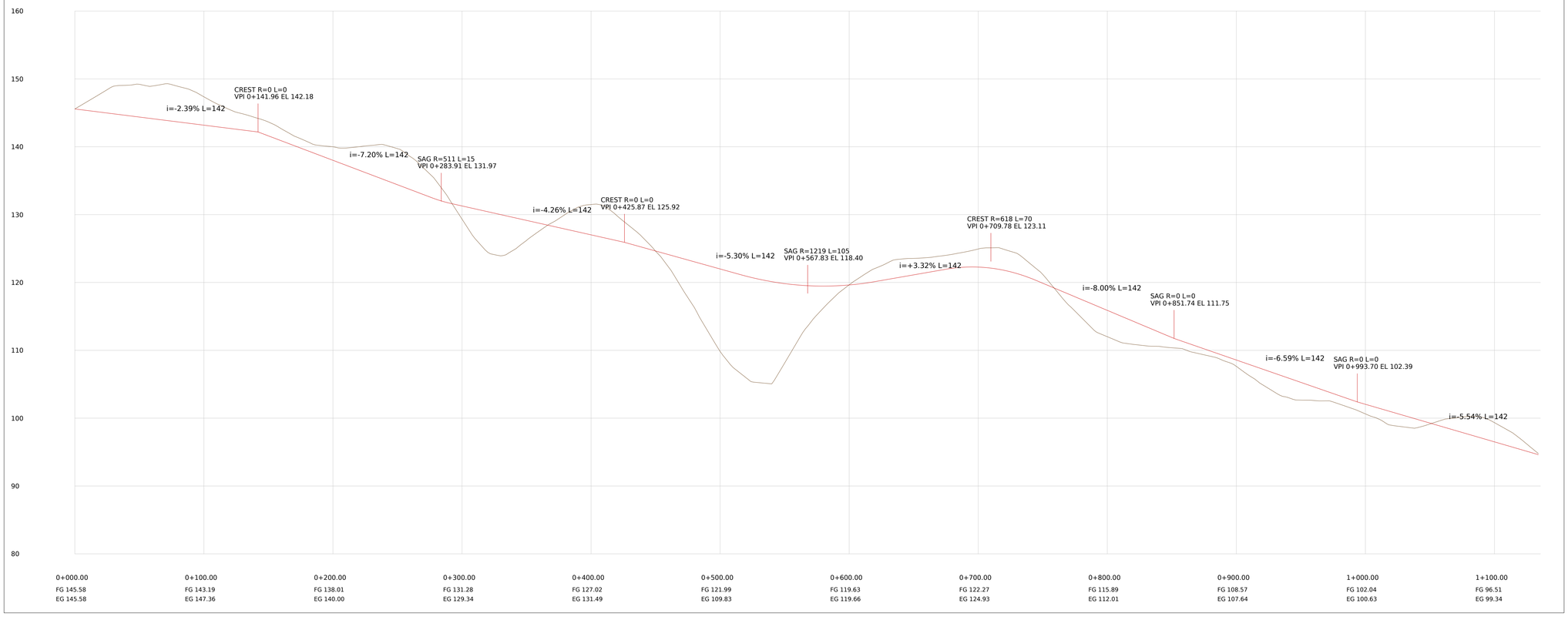
Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

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PLAN 1:4000 contours 1 m (index 5 m) N↑



PROFILE H 1:1829 V 1:348 (existing ground brown, finished grade red)



Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Plan and profile

Sheet 3 of 32

Scale: plan 1:4000; profile as shown (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

175bc80d01efaeeb29d3307375fe64e2

a486d4e6670d7de3a340c8953107d8d8

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

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PRELIMINARY

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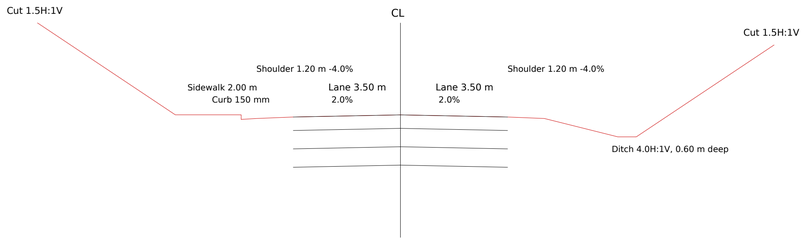
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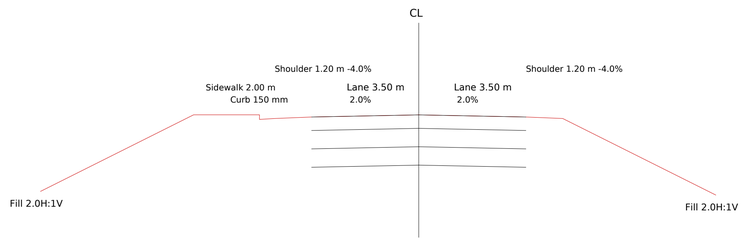
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TYPICAL SECTION IN CUT



TYPICAL SECTION IN FILL



Layer (top to bottom)	Thickness mm
Asphalt concrete	110
Aggregate base	150
Aggregate subbase	150

Pavement layer lines are drawn at 4x vertical exaggeration for legibility. Side slopes shown for a nominal 3.0 m cut / 2.5 m fill height; actual daylight per cross sections. Barrier, lining and protection treatments per the protection layout sheet.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Typical sections

Sheet 4 of 32

Scale: 1:100 (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

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Prepared by: civil-mcp agent tools

Engineer in responsible charge:

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PRELIMINARY

NOT FOR CONSTRUCTION

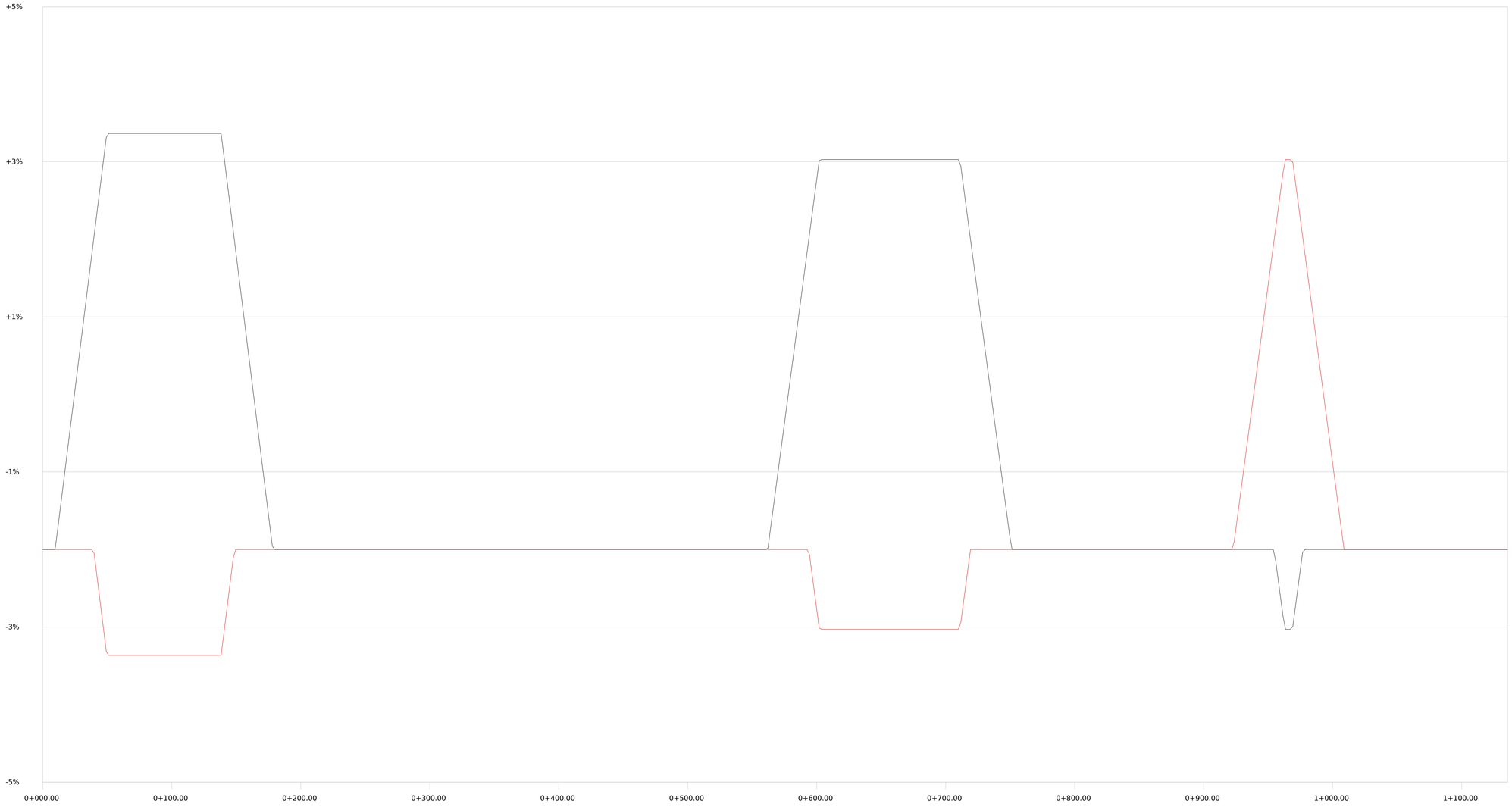
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Right edge cross slope
Left edge cross slope (rise per run outward from CL)



Curve	R m	e	Method	Runoff start	Full e	Full e end	Runoff end	Rel. gradient %
1	180	0.034	proportional distribution (AASHTO Method 1): e = emax·Rmin/R = 0.034·175.0...	0+049.58	0+049.58	0+138.25	0+178.25	0.469
2	200	0.030	proportional distribution (AASHTO Method 1): e = emax·Rmin/R = 0.030·175.0...	0+562.02	0+602.02	0+711.01	0+751.01	0.440
3	200	0.030	proportional distribution (AASHTO Method 1): e = emax·Rmin/R = 0.030·175.0...	0+922.83	0+962.83	0+968.80	1+008.80	0.440

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Superelevation diagram
Sheet 5 of 32
Scale: H 1:as shown; V as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
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Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

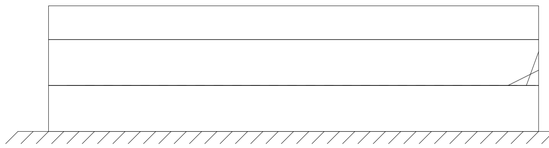
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PAVEMENT STRUCTURE (FLEXIBLE)



Asphalt concrete - 110 mm
a = 0.44, m = 1

Aggregate base - 150 mm
a = 0.14, m = 1

Aggregate subbase - 150 mm
a = 0.11, m = 1

Prepared subgrade, design CBR 6%

Total thickness 410 mm

DESIGN BASIS

Method: AASHTO 1993 flexible: structural number with layered analysis (AASHTO Guide for Design of Pavement Structures (1993), Part II Ch. 3 (flexible design equation, layered design analysis))
Structural number required 2.91, provided 3.38
- SN over subgrade (Mr = 9000 psi) = 2.91
- Asphalt concrete: SN required above the layer below = 1.87; D = (SN − ΣSN_above)/(a·m) = (1.87 − 0.00)/(0.44·1.00) = 4.24 in → 110 mm (min 75 mm); SN provided so far 1.91
- Aggregate base: SN required above the layer below = 2.43; D = (SN − ΣSN_above)/(a·m) = (2.43 − 1.91)/(0.14·1.00) = 3.73 in → 150 mm (min 150 mm); SN provided so far 2.73
- Aggregate subbase: SN required above the layer below = 2.91; D = (SN − ΣSN_above)/(a·m) = (2.91 − 2.73)/(0.11·1.00) = 1.64 in → 150 mm (min 150 mm); SN provided so far 3.38
Engineer inputs: reliability ZR; overall standard deviation S0; initial/terminal serviceability p0/pt; layer coefficients a_i, drainage m_i, layer moduli
Note: subgrade Mr from Mr(psi) = 1500·CBR (AASHTO 1993 correlation); confirm with tests

SUBGRADE PREPARATION (SCHEMATIC)



Strip topsoil and unsuitable material; proof-roll exposed subgrade.
Scarify and recompact the top 300 mm (dashed) to the specified density.
Replace soft spots identified by proof-rolling (engineer to direct).
Subgrade drains where the geotechnical report requires them.
Density, moisture and CBR testing per the governing specification.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Pavement and subgrade details
Sheet 6 of 32
Scale: 1:10 (layers) (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
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Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

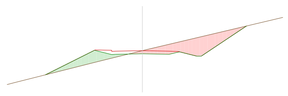
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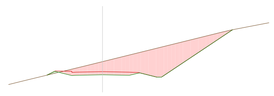
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STA 0+000.00



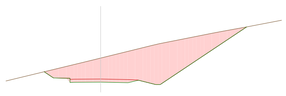
FG 145.58 EG 145.58 cut 18.3 m² fill 7.9 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 12.7 m

STA 0+009.58



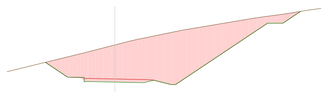
FG 145.35 EG 146.65 cut 45.0 m² fill 0.2 m²
cross slope L -2.00% R -2.00% daylight L 17.0 R 7.3 m

STA 0+020.00



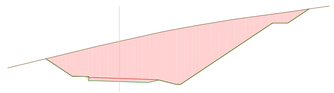
FG 145.10 EG 147.84 cut 77.0 m² fill 0.0 m²
cross slope L -2.00% R -0.60% daylight L 19.1 R 7.4 m

STA 0+040.00



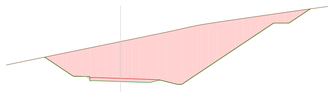
FG 144.62 EG 149.06 cut 124.3 m² fill 0.0 m²
cross slope L -2.08% R +2.08% daylight L 24.3 R 9.1 m

STA 0+049.58



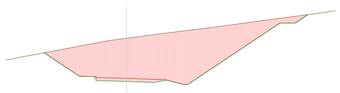
FG 144.39 EG 149.23 cut 137.5 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 24.8 R 9.6 m

STA 0+060.00



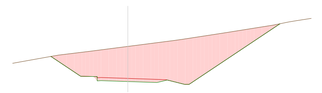
FG 144.14 EG 148.95 cut 137.9 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 24.8 R 9.9 m

STA 0+080.00



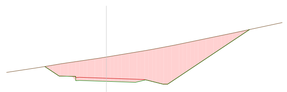
FG 143.66 EG 148.90 cut 137.4 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 23.6 R 10.8 m

STA 0+100.00



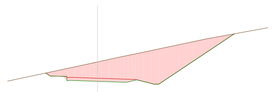
FG 143.19 EG 147.36 cut 107.1 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 19.9 R 10.0 m

STA 0+120.00



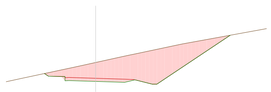
FG 142.71 EG 145.48 cut 74.1 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 18.7 R 8.0 m

STA 0+138.25



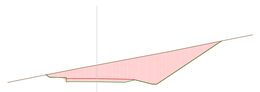
FG 142.27 EG 144.40 cut 60.7 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 17.9 R 6.8 m

STA 0+140.00



FG 142.23 EG 144.29 cut 58.9 m² fill 0.0 m²
cross slope L -3.13% R +3.13% daylight L 17.6 R 6.7 m

STA 0+160.00



FG 140.88 EG 142.65 cut 48.5 m² fill 0.0 m²
cross slope L -2.00% R +0.45% daylight L 16.3 R 6.6 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 7 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

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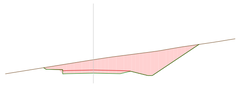
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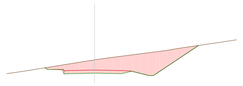
CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

STA 0+178.25



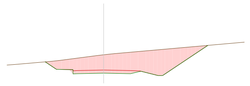
FG 139.57 EG 140.92 cut 34.4 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.8 R 6.5 m

STA 0+180.00



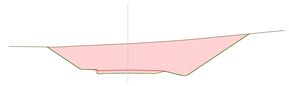
FG 139.45 EG 140.77 cut 33.4 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 6.5 m

STA 0+200.00



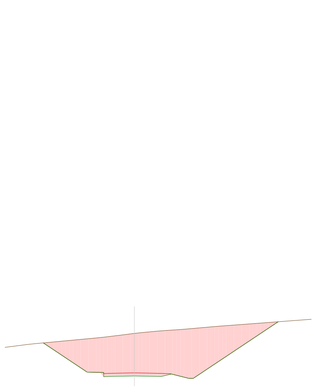
FG 138.01 EG 140.00 cut 43.9 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 7.6 m

STA 0+220.00



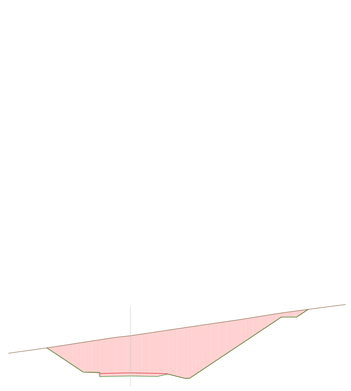
FG 136.57 EG 140.00 cut 79.3 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.9 R 10.5 m

STA 0+240.00



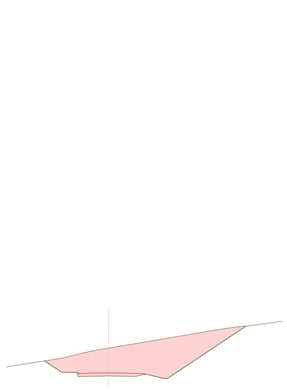
FG 135.13 EG 140.28 cut 125.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 18.8 R 11.9 m

STA 0+260.00



FG 133.69 EG 138.54 cut 128.7 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 23.2 R 10.9 m

STA 0+276.41



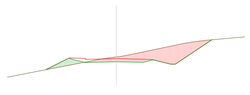
FG 132.50 EG 135.74 cut 80.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 18.0 R 8.4 m

STA 0+280.00



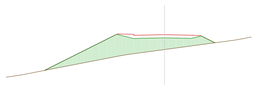
FG 132.26 EG 134.96 cut 66.9 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 16.7 R 7.7 m

STA 0+291.41



FG 131.65 EG 131.87 cut 18.5 m² fill 2.3 m²
cross slope L -2.00% R -2.00% daylight L 12.5 R 9.2 m

STA 0+300.00



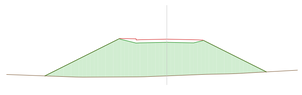
FG 131.28 EG 129.34 cut 0.0 m² fill 32.9 m²
cross slope L -2.00% R -2.00% daylight L 6.6 R 15.7 m

STA 0+320.00



FG 130.43 EG 124.44 cut 0.0 m² fill 142.7 m²
cross slope L -2.00% R -2.00% daylight L 12.2 R 25.7 m

STA 0+340.00



FG 129.58 EG 124.78 cut 0.0 m² fill 91.9 m²
cross slope L -2.00% R -2.00% daylight L 13.0 R 15.9 m

CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 8 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

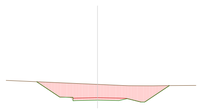
Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE

STA 0+360.00

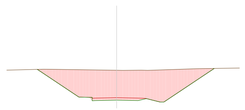
FG 128.72 EG 127.60 cut 0.0 m² fill 10.2 m²
cross slope L -2.00% R -2.00% daylight L 6.9 R 8.2 m

STA 0+380.00



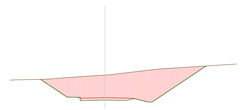
FG 127.87 EG 130.04 cut 44.6 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 11.7 R 9.9 m

STA 0+400.00



FG 127.02 EG 131.49 cut 104.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.9 R 12.9 m

STA 0+420.00



FG 126.17 EG 129.85 cut 88.4 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 16.5 R 10.5 m

STA 0+440.00

FG 125.17 EG 126.64 cut 36.0 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 6.8 m

STA 0+460.00

FG 124.11 EG 122.28 cut 0.0 m² fill 30.3 m²
cross slope L -2.00% R -2.00% daylight L 6.0 R 14.6 m

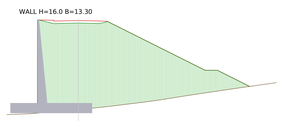
STA 0+480.00

FG 123.05 EG 116.32 cut 0.0 m² fill 184.3 m²
cross slope L -2.00% R -2.00% daylight L 14.0 R 29.4 m

STA 0+500.00

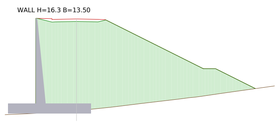
FG 121.99 EG 109.83 cut 0.0 m² fill 234.0 m²
cross slope L -2.00% R -2.00% daylight L 23.5 R 6.7 m

STA 0+515.33



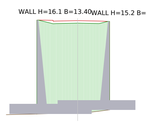
FG 121.18 EG 106.67 cut 0.0 m² fill 319.4 m²
cross slope L -2.00% R -2.00% daylight L 28.0 R 6.7 m

STA 0+520.00



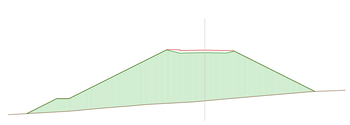
FG 120.94 EG 105.97 cut 0.0 m² fill 339.7 m²
cross slope L -2.00% R -2.00% daylight L 29.2 R 6.7 m

STA 0+540.00



FG 120.12 EG 105.05 cut 0.0 m² fill 174.1 m²
cross slope L -2.00% R -2.00% daylight L 5.2 R 6.7 m

STA 0+560.00



FG 119.63 EG 111.35 cut 0.0 m² fill 232.8 m²
cross slope L -2.00% R -2.00% daylight L 18.0 R 29.1 m

CROSS SECTIONS 1:500 (red = cut, green = fill, green line = subgrade datum)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 9 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

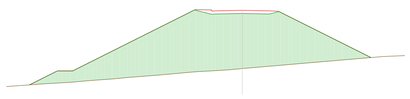
NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

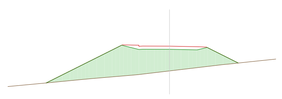
ENGINEER SEAL / SIGNATURE

STA 0+562.02



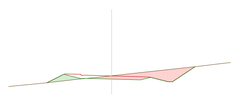
FG 119.60 EG 111.99 cut 0.0 m² fill 204.2 m²
cross slope L -2.00% R -2.00% daylight L 16.8 R 27.8 m

STA 0+580.00



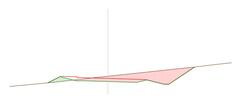
FG 119.47 EG 116.23 cut 0.0 m² fill 57.7 m²
cross slope L -2.00% R +0.26% daylight L 9.0 R 16.1 m

STA 0+600.00



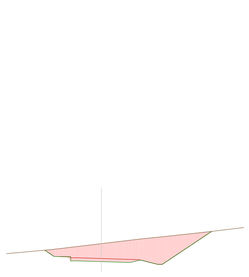
FG 119.63 EG 119.66 cut 10.4 m² fill 2.0 m²
cross slope L -2.77% R +2.77% daylight L 10.9 R 8.5 m

STA 0+602.02



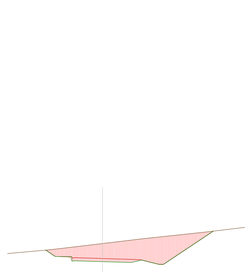
FG 119.67 EG 119.95 cut 14.3 m² fill 1.1 m²
cross slope L -3.03% R +3.03% daylight L 11.3 R 7.8 m

STA 0+620.00



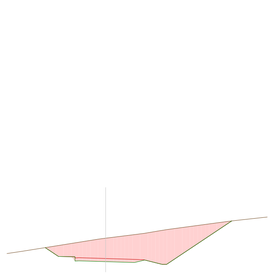
FG 120.13 EG 122.07 cut 45.7 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 14.4 R 7.4 m

STA 0+620.33



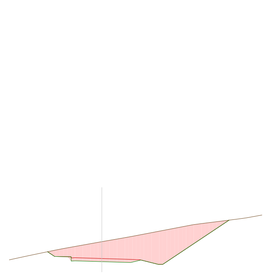
FG 120.14 EG 122.09 cut 46.1 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 14.5 R 7.4 m

STA 0+640.00



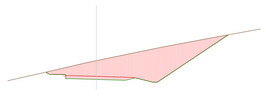
FG 120.79 EG 123.44 cut 64.0 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 16.5 R 7.9 m

STA 0+660.00



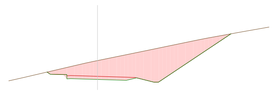
FG 121.45 EG 123.66 cut 58.1 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 16.6 R 7.1 m

STA 0+674.78



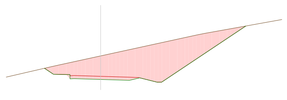
FG 121.95 EG 124.00 cut 56.6 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 17.2 R 6.6 m

STA 0+680.00



FG 122.10 EG 124.15 cut 57.4 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 17.4 R 6.6 m

STA 0+700.00



FG 122.27 EG 124.93 cut 74.7 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 19.0 R 7.3 m

STA 0+711.01



FG 122.09 EG 125.11 cut 85.1 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 19.9 R 7.7 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 10 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

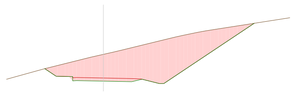
NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE

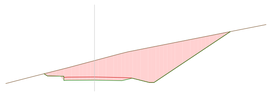
CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

STA 0+720.00



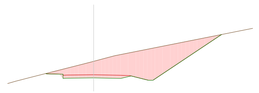
FG 121.79 EG 124.88 cut 86.7 m² fill 0.0 m²
cross slope L -2.00% R +1.90% daylight L 19.7 R 7.6 m

STA 0+740.00



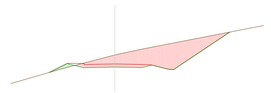
FG 120.67 EG 122.81 cut 60.7 m² fill 0.0 m²
cross slope L -2.00% R -0.62% daylight L 17.7 R 6.6 m

STA 0+744.78



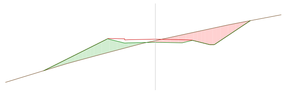
FG 120.31 EG 122.06 cut 50.4 m² fill 0.0 m²
cross slope L -2.00% R -1.22% daylight L 16.7 R 6.2 m

STA 0+751.01



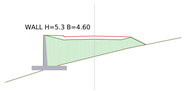
FG 119.81 EG 120.97 cut 36.1 m² fill 0.8 m²
cross slope L -2.00% R -2.00% daylight L 15.0 R 8.6 m

STA 0+760.00



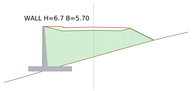
FG 119.09 EG 118.90 cut 13.7 m² fill 10.9 m²
cross slope L -2.00% R -2.00% daylight L 12.4 R 14.6 m

STA 0+780.00



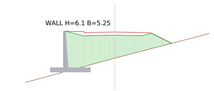
FG 117.49 EG 114.82 cut 0.0 m² fill 30.4 m²
cross slope L -2.00% R -2.00% daylight L 6.7 R 6.7 m

STA 0+800.00



FG 115.89 EG 112.01 cut 0.0 m² fill 46.9 m²
cross slope L -2.00% R -2.00% daylight L 7.9 R 6.7 m

STA 0+820.00



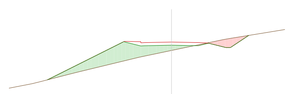
FG 114.29 EG 110.88 cut 0.0 m² fill 40.5 m²
cross slope L -2.00% R -2.00% daylight L 7.4 R 6.7 m

STA 0+840.00



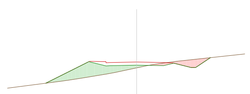
FG 112.69 EG 110.59 cut 0.0 m² fill 53.3 m²
cross slope L -2.00% R -2.00% daylight L 5.8 R 26.4 m

STA 0+860.00



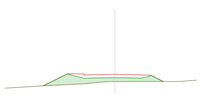
FG 111.21 EG 110.11 cut 3.6 m² fill 22.1 m²
cross slope L -2.00% R -2.00% daylight L 10.1 R 16.2 m

STA 0+880.00



FG 109.89 EG 109.12 cut 3.8 m² fill 12.4 m²
cross slope L -2.00% R -2.00% daylight L 9.6 R 11.9 m

STA 0+900.00



FG 108.57 EG 107.64 cut 0.0 m² fill 9.9 m²
cross slope L -2.00% R -2.00% daylight L 6.3 R 9.3 m

CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 11 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

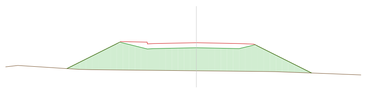
NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

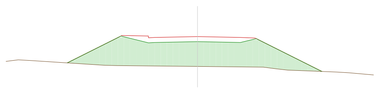
ENGINEER SEAL / SIGNATURE

STA 0+920.00



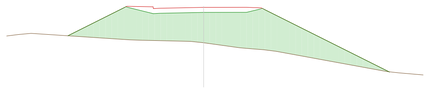
FG 107.25 EG 104.96 cut 0.0 m² fill 30.9 m²
cross slope L -2.00% R -2.00% daylight L 9.4 R 10.6 m

STA 0+922.83



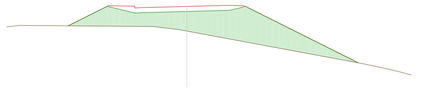
FG 107.06 EG 104.63 cut 0.0 m² fill 34.3 m²
cross slope L -2.00% R -2.00% daylight L 10.2 R 10.6 m

STA 0+940.00



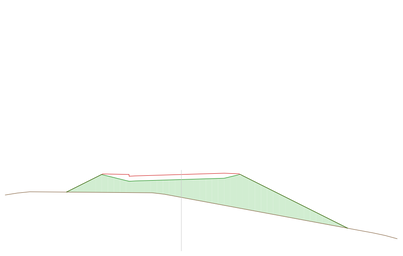
FG 105.93 EG 103.04 cut 0.0 m² fill 52.0 m²
cross slope L +0.16% R -2.00% daylight L 15.2 R 11.1 m

STA 0+960.00



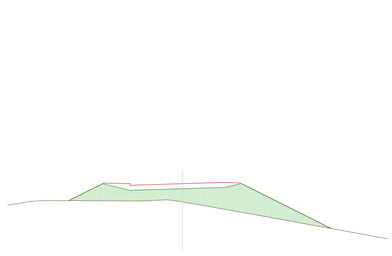
FG 104.61 EG 102.61 cut 0.0 m² fill 34.7 m²
cross slope L +2.67% R -2.67% daylight L 14.0 R 9.7 m

STA 0+962.83



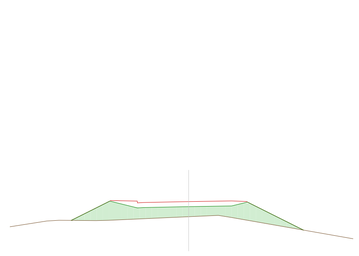
FG 104.43 EG 102.56 cut 0.0 m² fill 31.3 m²
cross slope L +3.03% R -3.03% daylight L 13.6 R 9.4 m

STA 0+968.80



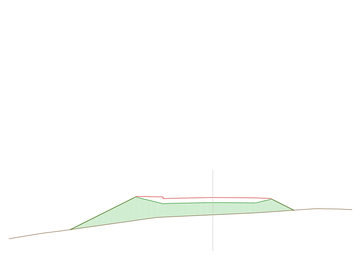
FG 104.03 EG 102.55 cut 0.0 m² fill 24.6 m²
cross slope L +3.03% R -3.03% daylight L 12.1 R 9.3 m

STA 0+980.00



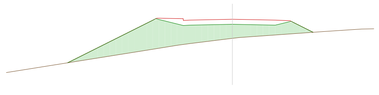
FG 103.30 EG 102.06 cut 0.0 m² fill 16.7 m²
cross slope L +1.62% R -2.00% daylight L 9.4 R 9.6 m

STA 1+000.00



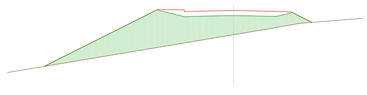
FG 102.04 EG 100.63 cut 0.0 m² fill 18.1 m²
cross slope L -0.89% R -2.00% daylight L 6.6 R 11.6 m

STA 1+008.80



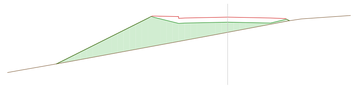
FG 101.56 EG 100.00 cut 0.0 m² fill 24.2 m²
cross slope L -2.00% R -2.00% daylight L 6.6 R 13.4 m

STA 1+020.00



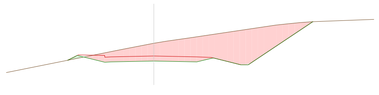
FG 100.94 EG 98.96 cut 0.0 m² fill 33.8 m²
cross slope L -2.00% R -2.00% daylight L 6.4 R 15.5 m

STA 1+040.00



FG 99.83 EG 98.59 cut 0.0 m² fill 20.4 m²
cross slope L -2.00% R -2.00% daylight L 5.0 R 14.0 m

STA 1+060.00



FG 98.72 EG 99.76 cut 28.9 m² fill 0.2 m²
cross slope L -2.00% R -2.00% daylight L 13.0 R 7.0 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 12 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

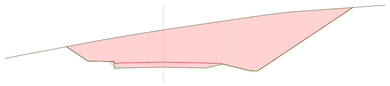
NOT FOR CONSTRUCTION

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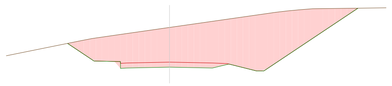
CROSS SECTIONS 1:250 (red = cut, green = fill, green line = subgrade datum)

STA 1+080.00



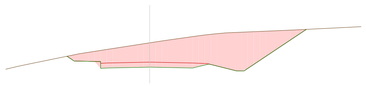
FG 97.61 EG 100.25 cut 61.9 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.5 R 7.9 m

STA 1+100.00



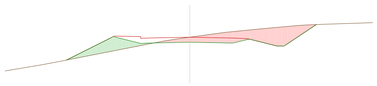
FG 96.51 EG 99.34 cut 66.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.4 R 8.3 m

STA 1+120.00



FG 95.40 EG 96.98 cut 36.2 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 12.8 R 6.7 m

STA 1+135.65



FG 94.53 EG 94.53 cut 9.8 m² fill 3.8 m²
cross slope L -2.00% R -2.00% daylight L 10.3 R 10.1 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 13 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

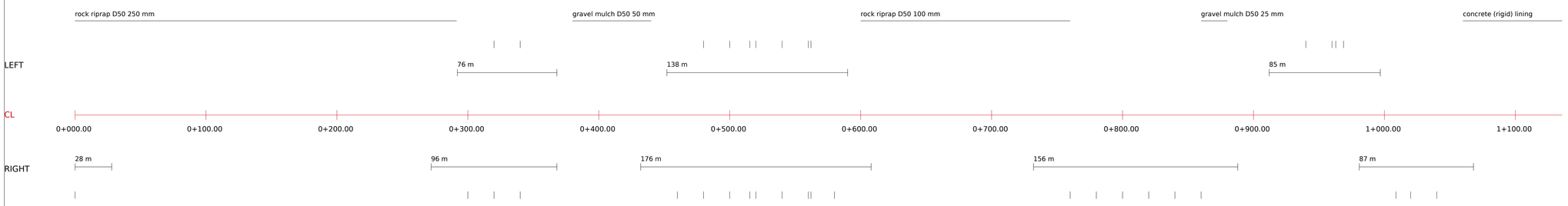
NOT FOR CONSTRUCTION

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CROSS SECTIONS 1:250 (red = cut, green = fill, green line = subgrade datum)

Ticks: hazards inside the clear zone (embankment above warrant height, steep slopes, walls). Thick lines: barrier runs incl. length of need. Blue: ditch lining reaches.



Side	From	To	Length m	LON m	Type	Reason
left	0+292.00	0+368.00	76	28.0	W-beam guardrail	critical fill slope
left	0+452.00	0+590.02	138	28.0	W-beam guardrail	critical fill slope; retaining wall drop-off
left	0+912.00	0+996.80	85	28.0	W-beam guardrail	critical fill slope
right	0+000.00	0+028.00	28	28.0	W-beam guardrail	critical fill slope
right	0+272.00	0+368.00	96	28.0	W-beam guardrail	critical fill slope
right	0+432.00	0+608.00	176	28.0	W-beam guardrail	critical fill slope; retaining wall drop-off
right	0+732.00	0+888.00	156	28.0	W-beam guardrail	critical fill slope; retaining wall drop-off
right	0+980.80	1+068.00	87	28.0	W-beam guardrail	critical fill slope

Side	From	To	Q m3/s	Depth m	V m/s	Slope %	tau_d N/m2	Lining	tau_p N/m2
left	0+000.00	0+291.41	0.000	0.00	1.49	7277.59	221.3	rock riprap D50 250 mm	297
left	0+380.00	0+440.00	0.017	0.05	0.45	5.30	26.1	gravel mulch D50 50 mm	38
left	0+600.00	0+760.00	0.044	0.08	0.67	8.00	63.5	rock riprap D50 100 mm	76
left	0+860.00	0+880.00	0.006	0.02	0.36	6.59	15.0	gravel mulch D50 25 mm	19
left	1+060.00	1+135.65	0.021	0.00	11.13	4729.39	1439.2	concrete (rigid) lining	rigid

Clear zone 5 m (engineer input from AASHTO Roadside Design Guide Table 3-1 for the design speed, ADT and slopes (demo value)); recoverable ≤ 1V:4H, traversable ≤ 1V:3H, embankment warrant height 3 m: engineer inputs.
right 10: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
left 300: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
left 360: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
right 360: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
left 460: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Roadside protection layout (straight-line diagram)
Sheet 14 of 32
Scale: H 1:1800 (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

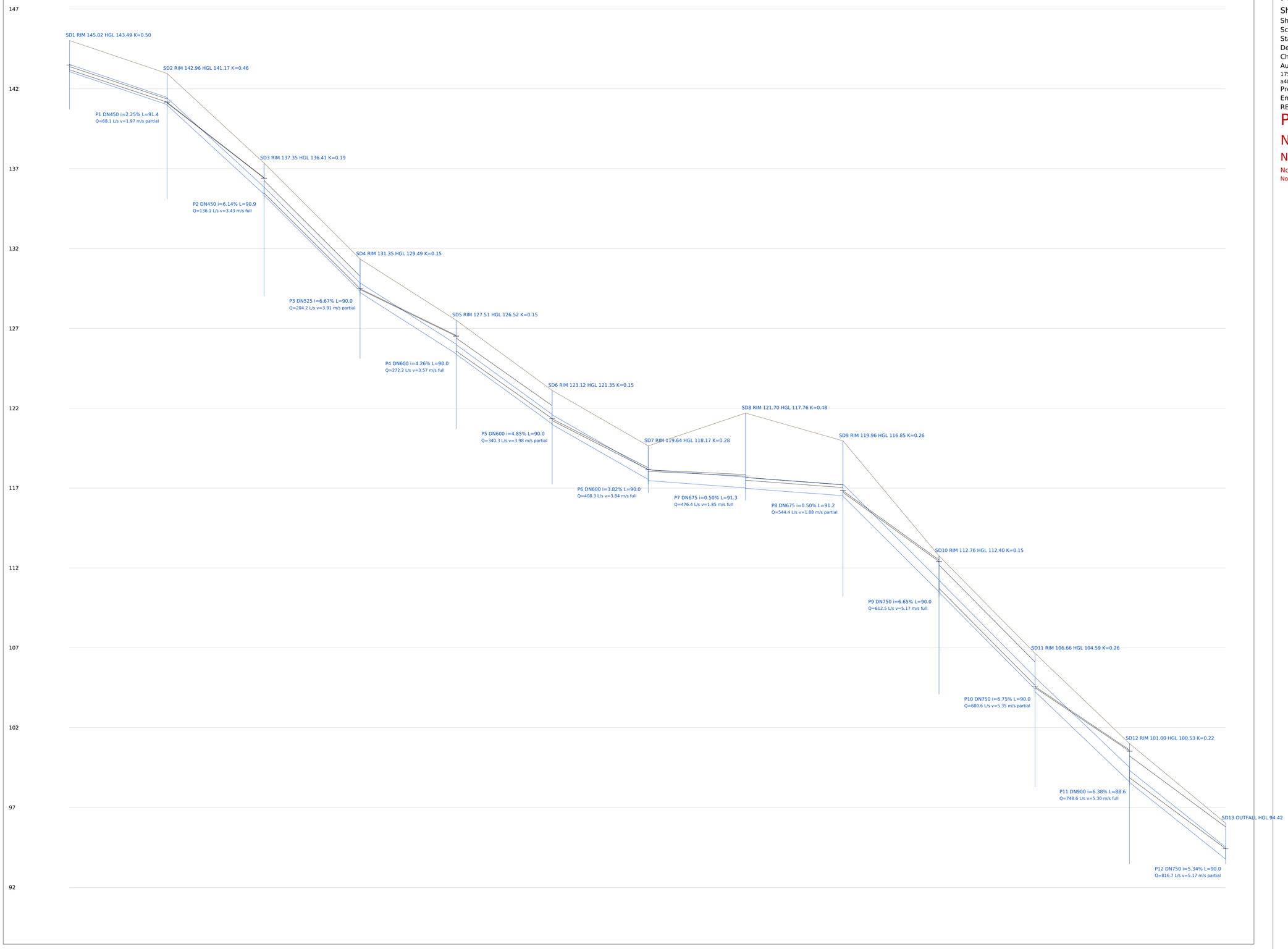
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PIPE PROFILE - Storm H 1:1773 V 1:119

cyan = HGL, red = EGL, brown = rim; method FHWA HEC-22 approximate method: outfall HGL = max(tailwater, invert + (dc+D)/2); full flow: HGL_up = HGL_dn + Sf·L, Sf = (Qn/(A R^(2/3)))²; partial flow: normal depth; structure: E = EGL_out + K·Vo²/2g, K by deflection (0°0.15, 22.5°0.45, 45°0.75, 60°0.85, 90°1.0; start structure 0.5). Coefficients transcribed — verify against the HEC-22 edition in use.



P-SURCH advisory pipe P2 Pipe flows full (surcharged): 0.554 vs max 0.000 m
P-SURCH advisory pipe P4 Pipe flows full (surcharged): 0.505 vs max 0.000 m
P-SURCH advisory pipe P6 Pipe flows full (surcharged): 0.023 vs max 0.000 m
P-SURCH advisory pipe P7 Pipe flows full (surcharged): 0.070 vs max 0.000 m
P-SURCH advisory pipe P9 Pipe flows full (surcharged): 1.146 vs max 0.000 m
P-SURCH advisory pipe P11 Pipe flows full (surcharged): 1.022 vs max 0.000 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Pipe profile Storm
Sheet 15 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

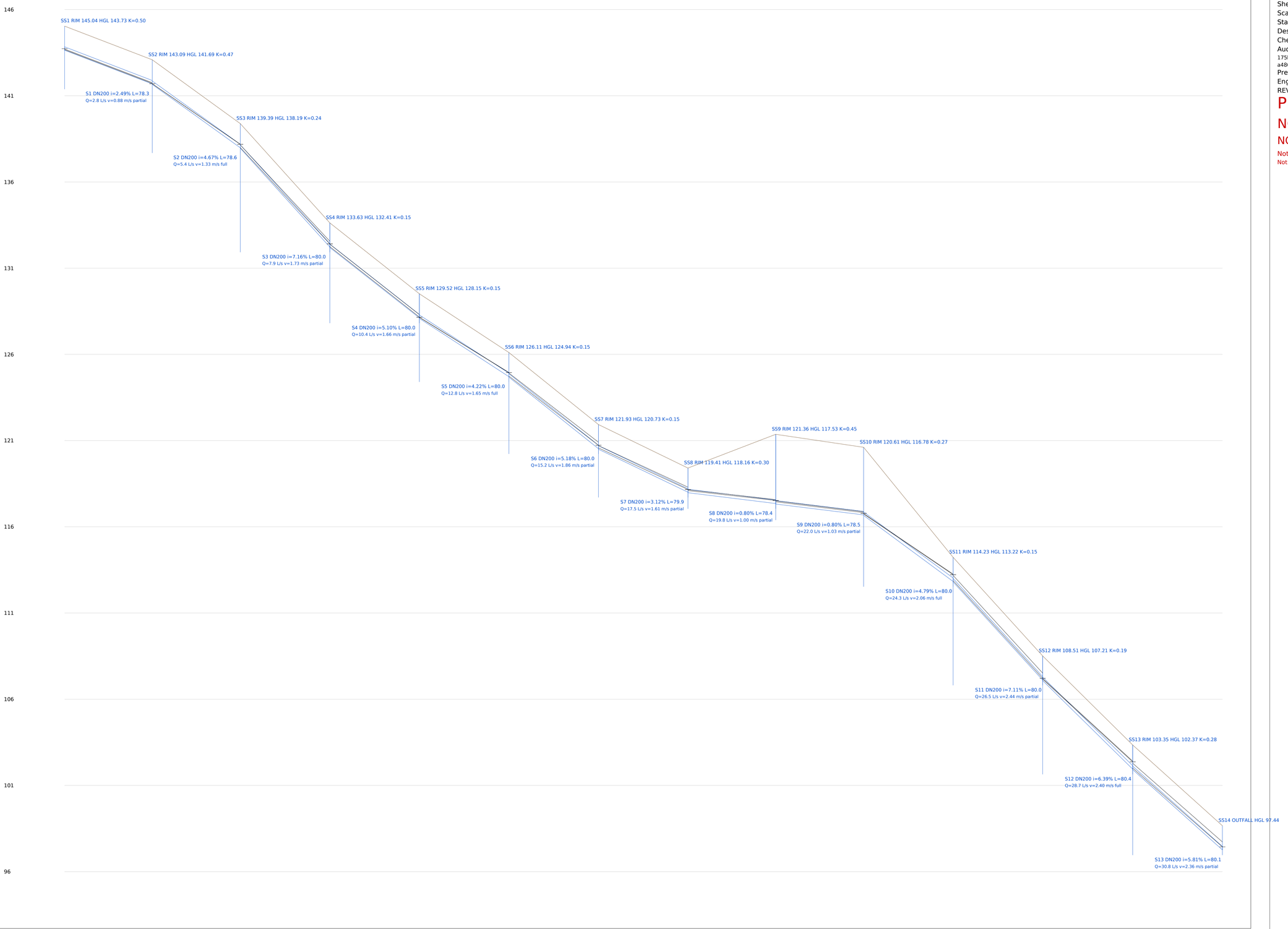
NOT FOR CONSTRUCTION

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PIPE PROFILE - Sanitary H 1:1693 V 1:110

cyan = HGL, red = EGL, brown = rim; method FHWA HEC-22 approximate method: outfall HGL = max(tailwater, invert + (dc+D)/2); full flow: HGL_up = HGL_dn + Sf·L, Sf = (Qn/(A R^(2/3)))²; partial flow: normal depth; structure: E = EGL_out + K·Vo²/2g, K by deflection (0°0.15, 22.5°0.45, 45°0.75, 60°0.85, 90°1.0; start structure 0.5). Coefficients transcribed — verify against the HEC-22 edition in use.



P-SURCH advisory pipe S2 Pipe flows full (surcharged): 0.001 vs max 0.000 m
P-SURCH advisory pipe S5 Pipe flows full (surcharged): 0.035 vs max 0.000 m
P-SURCH advisory pipe S10 Pipe flows full (surcharged): 0.195 vs max 0.000 m
P-SURCH advisory pipe S12 Pipe flows full (surcharged): 0.221 vs max 0.000 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Pipe profile Sanitary
Sheet 16 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

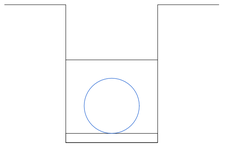
NOT FOR CONSTRUCTION

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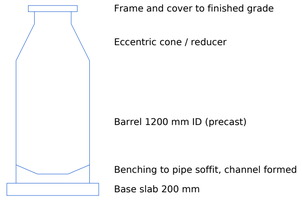
ENGINEER SEAL / SIGNATURE

PIPE TRENCH (SCHEMATIC, 1:50)

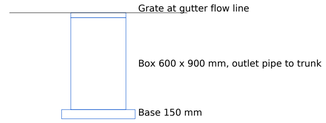


Trench width D + 600 mm = 1.50 m

MANHOLE (SCHEMATIC, 1:50)



GRATE INLET (SCHEMATIC, 1:50)



Network	Structure	Rim	Sump invert	Depth m	Inverts out (pipe, D mm)
Storm	SD1	145.024	143.074	1.95	P1 143.074 D450
Storm	SD2	142.963	140.983	1.98	P2 140.983 D450
Storm	SD3	137.352	135.327	2.03	P3 135.327 D525
Storm	SD4	131.346	129.246	2.10	P4 129.246 D600
Storm	SD5	127.512	125.382	2.13	P5 125.382 D600
Storm	SD6	123.116	120.986	2.13	P6 120.986 D600
Storm	SD7	119.643	117.468	2.17	P7 117.468 D675
Storm	SD8	121.698	116.981	4.72	P8 116.981 D675
Storm	SD9	119.961	116.495	3.47	P9 116.495 D750
Storm	SD10	112.756	110.476	2.28	P10 110.476 D750
Storm	SD11	106.655	104.255	2.40	P11 104.255 D900
Storm	SD12	101.004	98.574	2.43	P12 98.574 D750
Storm	SD13	96.020	93.770	2.25	
Sanitary	SS1	145.038	143.638	1.40	S1 143.638 D200
Sanitary	SS2	143.085	141.655	1.43	S2 141.655 D200
Sanitary	SS3	139.385	137.955	1.43	S3 137.955 D200
Sanitary	SS4	133.626	132.196	1.43	S4 132.196 D200
Sanitary	SS5	129.515	128.085	1.43	S5 128.085 D200
Sanitary	SS6	126.107	124.677	1.43	S6 124.677 D200
Sanitary	SS7	121.929	120.499	1.43	S7 120.499 D200
Sanitary	SS8	119.407	117.977	1.43	S8 117.977 D200
Sanitary	SS9	121.364	117.320	4.04	S9 117.320 D200
Sanitary	SS10	120.611	116.662	3.95	S10 116.662 D200
Sanitary	SS11	114.230	112.800	1.43	S11 112.800 D200
Sanitary	SS12	108.509	107.079	1.43	S12 107.079 D200
Sanitary	SS13	103.345	101.915	1.43	S13 101.915 D200
Sanitary	SS14	98.662	97.262	1.40	

Details are schematic. Final dimensions, materials, frames, grates, joint types and bedding class per the agency standard drawings and the governing specification cited on the standard-details sheet. Structure schedule lists every structure designed by the pipe-network tool (first 39 rows).

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Drainage details and structure schedule

Sheet 17 of 32

Scale: 1:50 (details) (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

175bc80d01efaeeb29d3307375fe64e2

a486d4e6670d7de3a340c8953107d8d8

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

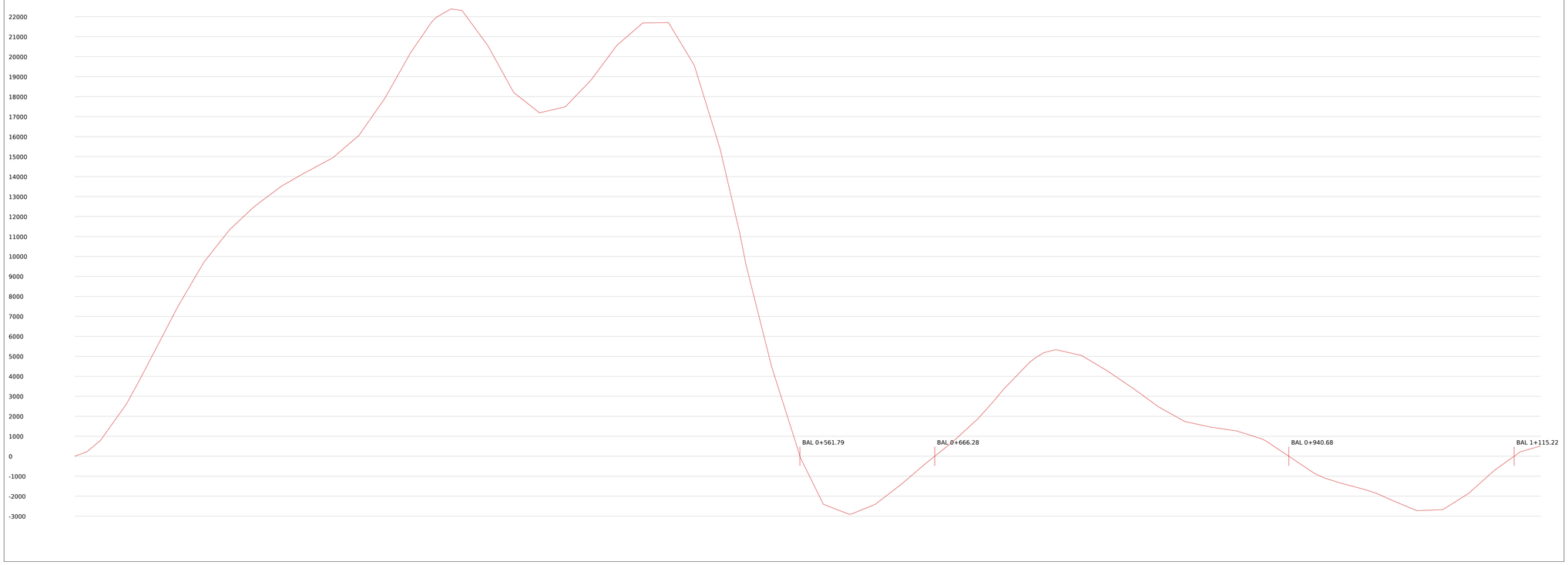
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MASS HAUL (m³) cut factor 0.90 final ordinate 514 m³



Station range	Cut m³	Fill m³	Asphalt concrete m³	Aggregate base m³	Aggregate subbase m³
0+000.00–0+100.00	10838	40	84.9	115.7	115.7
0+100.00–0+200.00	5811	0	84.4	115.2	115.2
0+200.00–0+300.00	8370	164	82.5	112.5	112.5
0+300.00–0+400.00	1936	5224	82.5	112.5	112.5
0+400.00–0+500.00	3533	6632	82.5	112.5	112.5
0+500.00–0+600.00	104	18378	83.0	113.2	113.2
0+600.00–0+700.00	5350	13	85.4	116.4	116.4
0+700.00–0+800.00	4022	1242	83.4	113.7	113.7
0+800.00–0+900.00	148	3135	82.5	112.5	112.5
0+900.00–1+000.00	0	2947	83.8	114.3	114.3
1+000.00–1+100.00	2480	1258	82.5	112.5	112.5
1+100.00–1+200.00	1386	30	29.4	40.1	40.1
Total	43976	39064	946.8	1291.1	1291.1

Pipe schedule

RCP DN450: 182.3 m
RCP DN525: 90.0 m
RCP DN600: 270.0 m
RCP DN675: 182.5 m
RCP DN750: 270.0 m
RCP DN900: 88.6 m
Structures: 13

Pipe schedule

PVC DN200: 1034.3 m
Structures: 14

Grading yard: cut 275 m³ fill 1006 m³ (grid 0.25 m)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Mass haul and quantities
Sheet 18 of 32
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

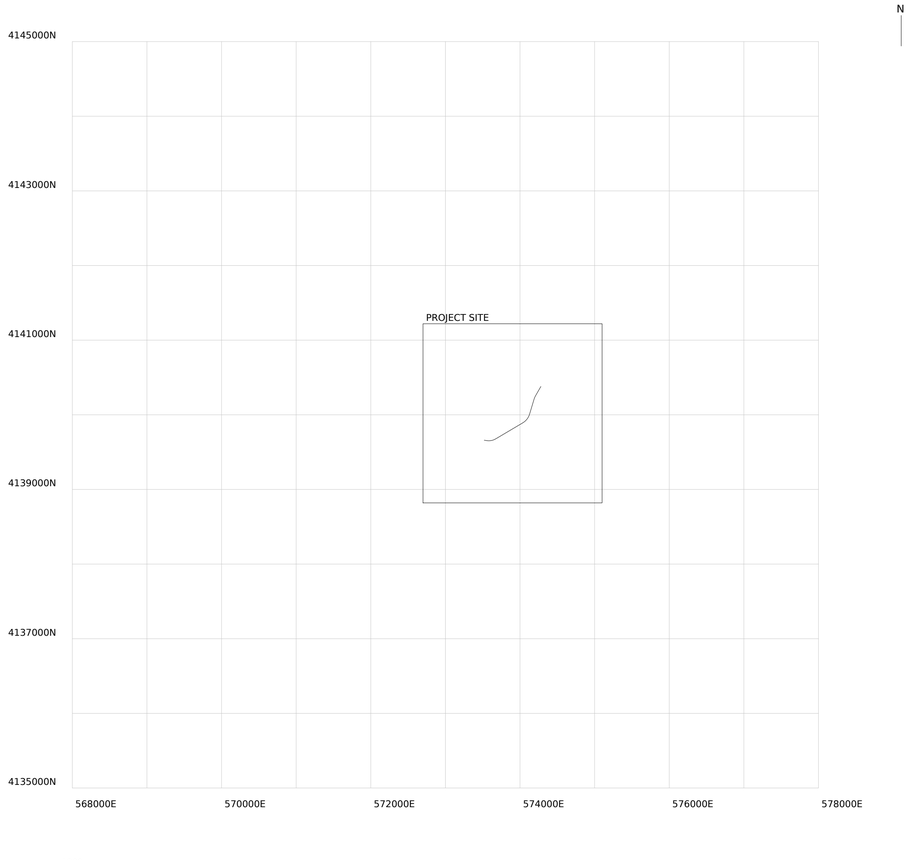
NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

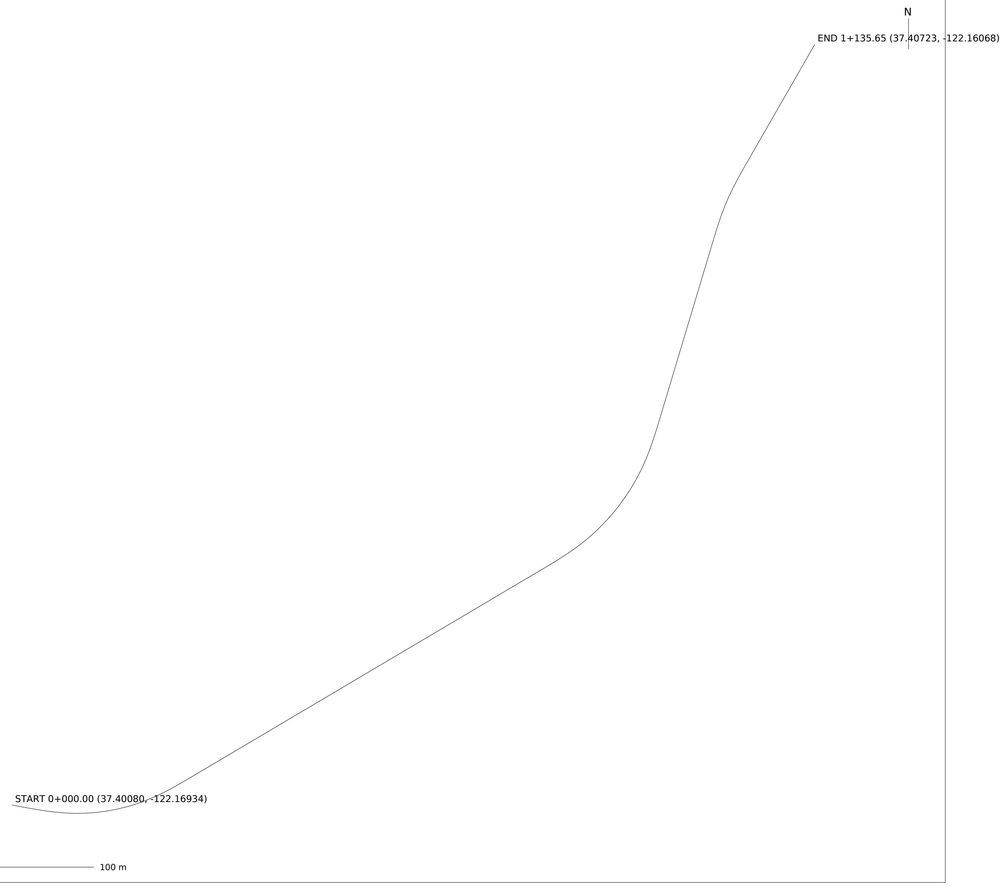
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Locality plan (UTM 1 km grid)



Project extent



Point	Easting	Northing	Latitude	Longitude
Start	573520.000	4139660.000	37.400803	-122.169338
End	574280.000	4140380.000	37.407232	-122.160679

WGS84 / UTM zone 10N (inverse transverse Mercator series, Snyder 1987, USGS PP 1395). No base mapping is shipped: overlay agency / OS / LINZ / USGS mapping under the user's licence.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Locality plan
Sheet 19 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

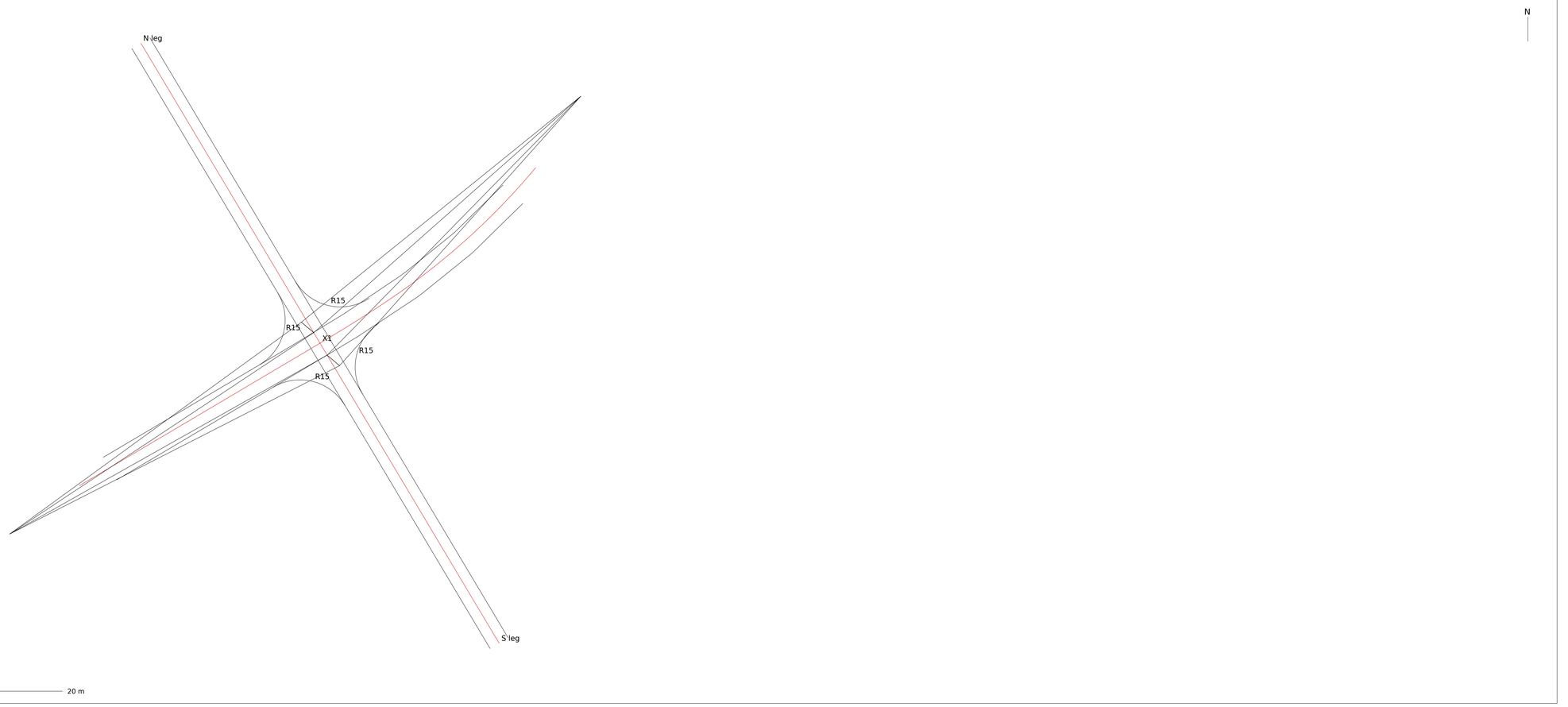
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Intersection layout



Intersection	Station	Required m	Avail. ahead m	Avail. back m
X1	0+570.00	104.3	153.0	468.0

ISD = 0.278*V*tg = 0.278*50*7.5. AASHTO Green Book 2018 §9.5.3 Case B1, time gap (AASHTO 2018 Table 9-6 passenger car left turn from stop (verify)). Available distance: vertical sight line along the major-road profile (eye 1.08 m, object 1.08 m), limited by the profile ends; horizontal obstructions in the sight triangle are cleared on site.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Intersection layout and sight distance

Sheet 20 of 32

Scale: as shown (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

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a486d4e6670d7de3a340c8953107d8d8

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

PRELIMINARY

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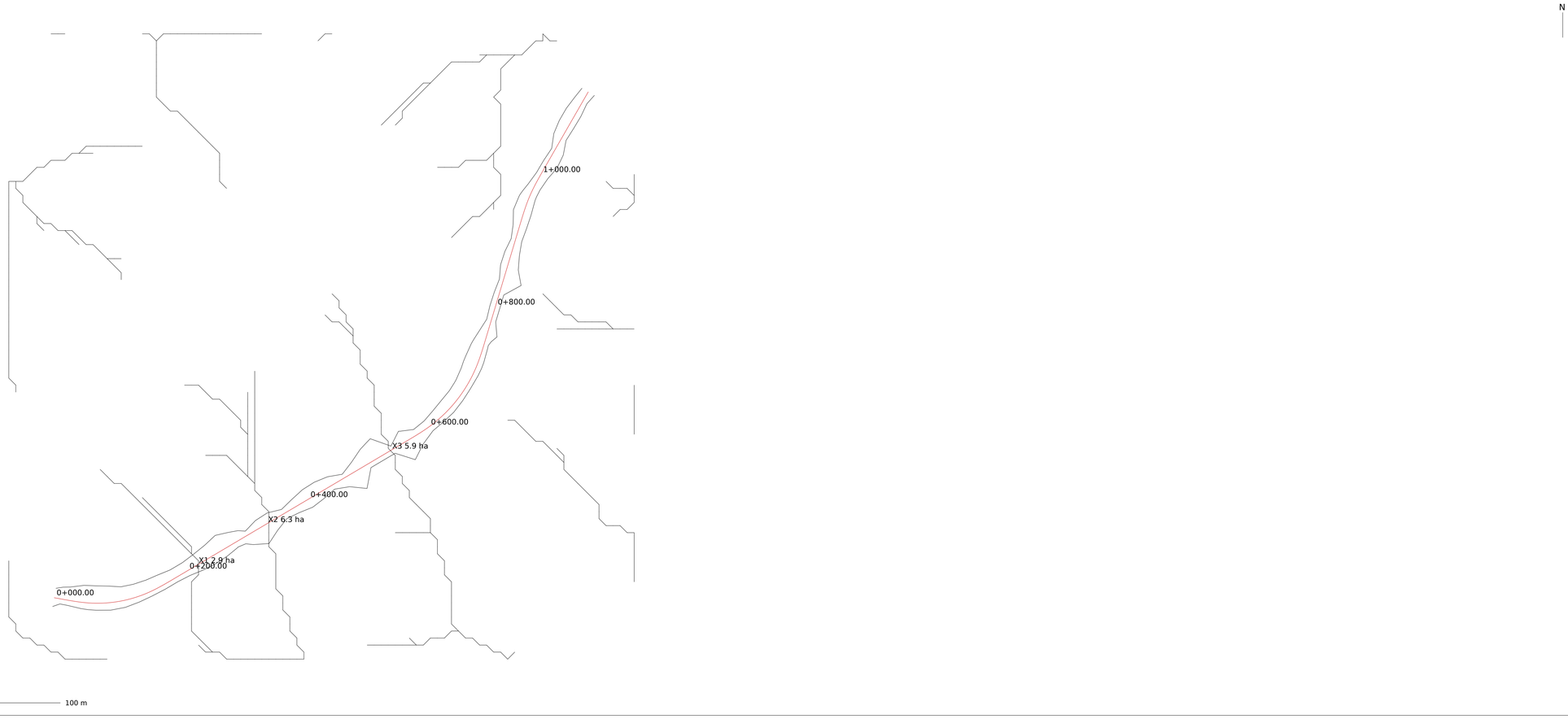
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ENGINEER SEAL / SIGNATURE

Drainage area map (D8 flow paths >= 0.5 ha, crossings)



ID	Sta	A ha	L m	S	tc min	i mm/h	Q m3/s	Method
X1	0+215.00	2.86	336	0.0646	5.0	86.0	0.239	rational
X2	0+330.00	6.34	414	0.0887	5.1	85.0	0.524	rational
X3	0+535.00	5.94	451	0.0953	5.3	83.4	0.482	rational

Method: Rational method Q = C·i·A (FHWA HDS-2 / HEC-22 §3.2). IFD: 10-yr illustrative IDF (DEMO VALUES - not NOAA Atlas 14; engineer to enter the PFDS table for the site). Runoff coefficient / curve number and IFD are engineer inputs.
Catchments from priority-flood + D8 on the survey TIN; confirm by site inspection and mapping. Area outside the survey extent is not captured.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Drainage area map and hydrology data
Sheet 21 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

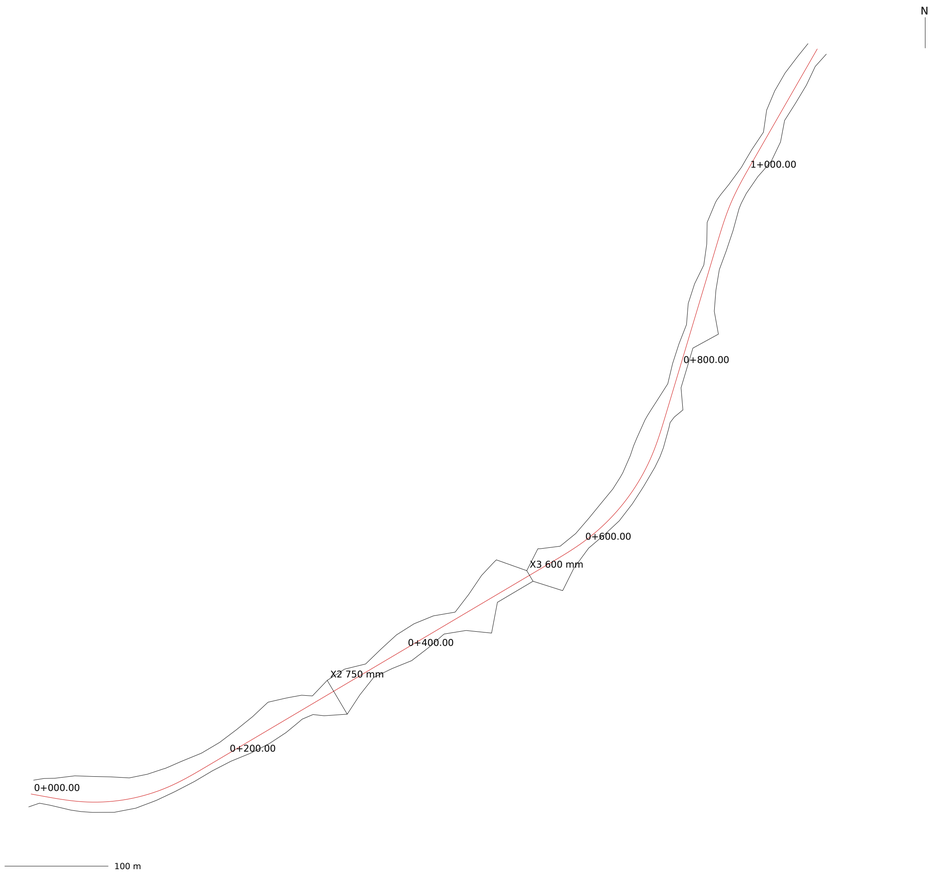
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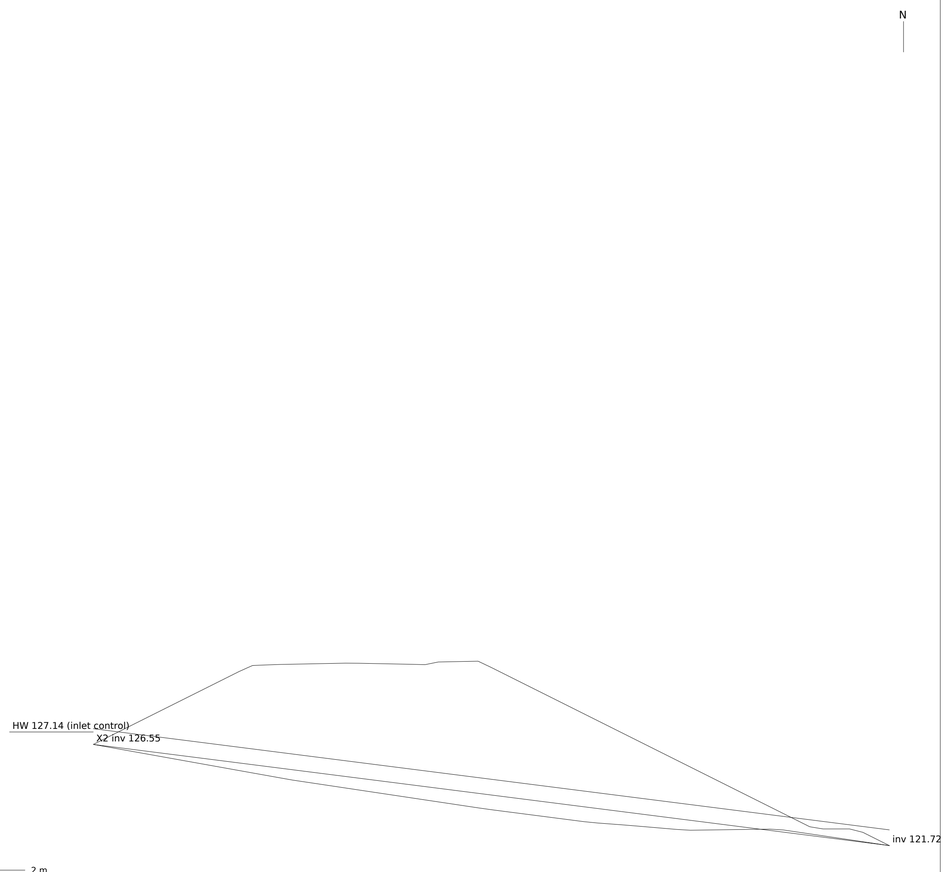
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Culvert layout



Longitudinal section, largest-flow culvert (V = H)



ID	Sta	Q m3/s	D m	L m	S %	Inv in	Inv out	HWi m	HWo m	HW/D	Ctrl	V out	Cover m
X1	0+215.00	0.239	-	-	-	-	-	-	-	-	in cut	-	-
X2	0+330.00	0.524	0.75	38.0	12.71	126.55	121.72	0.59	-4.07	0.79	inlet	6.59	4.17
X3	0+535.00	0.482	0.60	11.9	5.77	105.57	104.88	0.70	0.08	1.17	inlet	4.95	13.74

FHWA HDS-5 (3rd ed.): inlet control unsubmerged Form 1 for Q/(A D^0.5) <= 3.5 (Ku 1.811 SI), submerged for >= 4.0, linear between; outlet control full-flow energy with ho = max(TW, (dc+D)/2). Inlet ConcreteGrooveHeadwall, n 0.012, TW 0 m. Sizes, HW/D, freeboard and cover are engineer inputs (DEMO VALUE - engineer to replace).
Structural class (fill height, bedding, end treatment) per agency standard drawings; culverts above the agency structure threshold go to the bridge/structures hand-off.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Culvert layout, section and schedule
Sheet 22 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

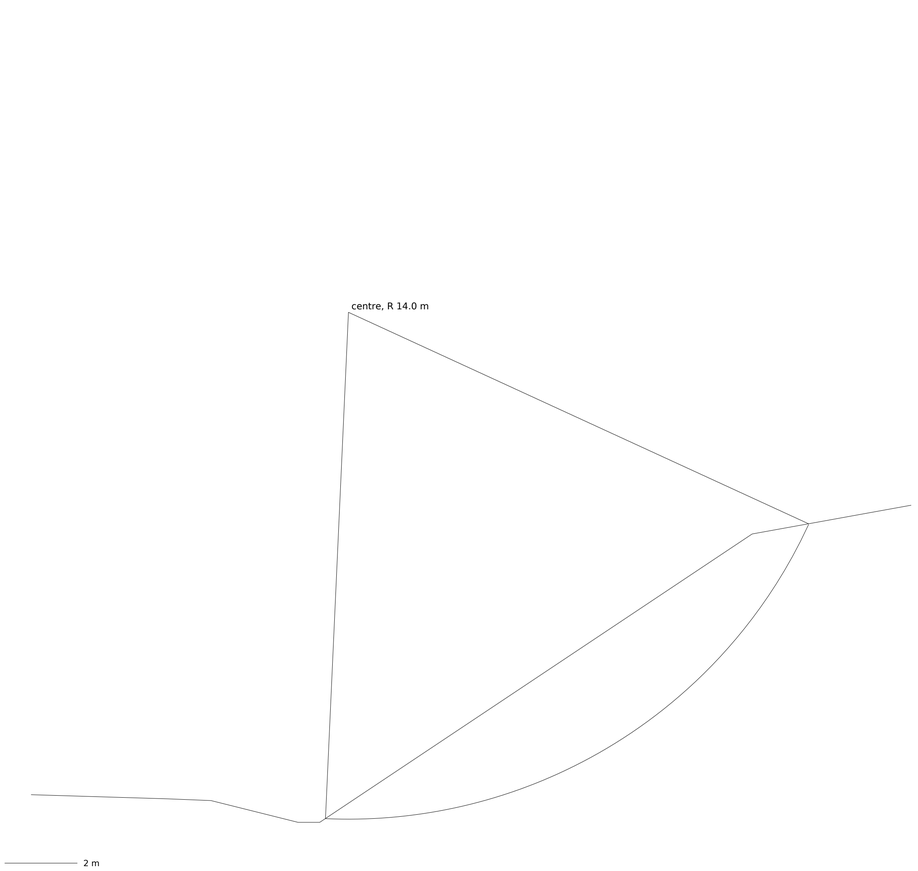
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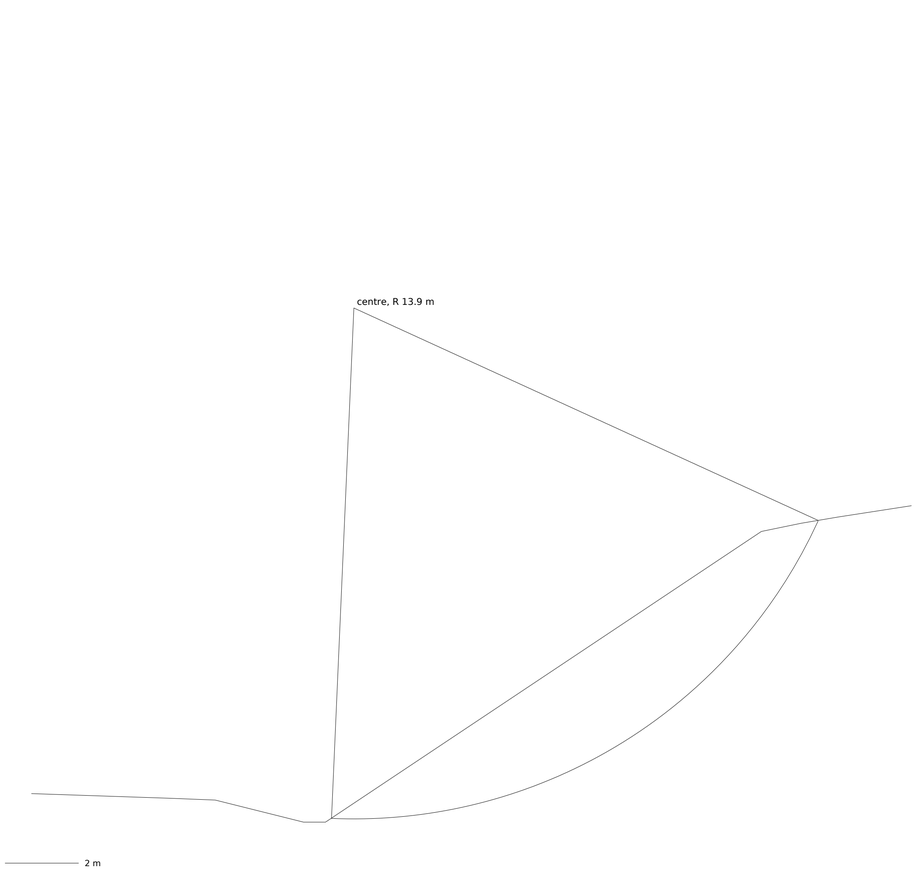
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Not reviewed or sealed by a licensed engineer.

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cut 0+711.01 L: FS 1.44 Bishop / 1.43 Spencer



cut 0+100.00 L: FS 1.44 Bishop / 1.44 Spencer



Sta	Side	Type	H m	FS Bishop	FS Spencer	theta deg	FS min	Xc	Zc	R
0+711.01	L	cut	8.0	1.438	1.434	26.1	1.30	8.8	135.4	14.0
0+100.00	L	cut	7.9	1.439	1.435	26.2	1.30	8.8	156.4	13.9
0+020.00	L	cut	7.5	1.479	1.474	25.8	1.30	9.1	156.0	11.7
0+560.00	L	fill	6.6	1.853	1.850	20.9	1.30	7.1	127.5	14.8
0+940.00	L	fill	5.2	2.006	2.002	20.1	1.30	6.7	112.3	11.9
0+860.00	R	fill	5.0	2.027	2.024	19.6	1.30	6.6	116.9	11.2

Soil c' 8 kPa, phi' 30 deg, gamma 19 kN/m3, ru 0.1 from GEOTECH REPORT PLACEHOLDER (no site investigation in the demo) (DEMO VALUE - engineer to replace). Homogeneous soil, circular surfaces, pore pressure by ru; Bishop simplified and Spencer (constant interslice inclination). Layering, water tables, seismic and reinforcement need the geotechnical designer's analysis using slope-sections.csv.
Criterion FS >= 1.3 (engineer / agency).

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Slope stability - critical slip circles
Sheet 23 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

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Edge drain detail (schematic, NTS)



Item	Value
qi (Cedergren)	0.2734 m3/day/m2
inflow per m	0.4785 m3/day/m
layer capacity	2.4739 m3/day/m
resultant slope / path	0.0825 / 7.22 m
t50	0.75 h
edge drain capacity	0.0158 m3/s
outlet spacing	150 m
AASHTO m (engineer)	1

qi = Ic(Nc/W + Wc/(W·Cs)) + kp = 0.223(2/1.75 + 1.75/(1.75·12.0)) + 0.0000 = 0.2734 m³/day/m²
inflow per m = qi·W = 0.4785 m³/day/m; layer capacity k·S_R·H = 2.4739 m³/day/m
resultant slope S_R = 0.0825, flow path L_R = 7.22 m, t50 = ne·L_R²/(2k(H+L_R·S_R)) = 0.75 h (approximation)
edge drain D100 mm full capacity 0.01583 m³/s; outlet spacing = min(Qp/inflow, max) = 150 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Subsurface (pavement) drainage
Sheet 24 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

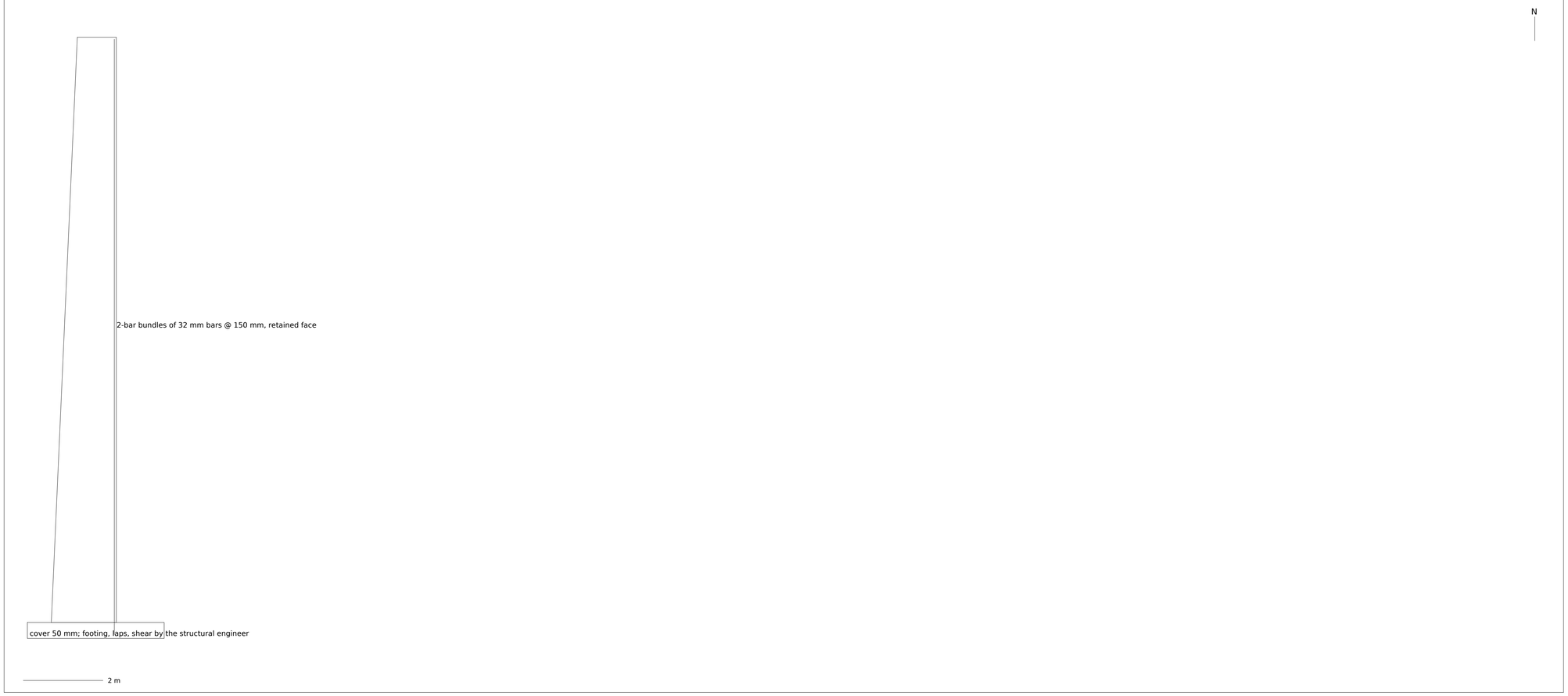
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Stem reinforcement, governing wall 0+520.00 R (schematic)



Sta	Side	Stem H m	t m	Mu kNm/m	d m	As req	As min	As mm2/m	Bar mm	Spacing mm	OK
0+540.00	L	13.71	1.52	4551.4	1.457	8722	2742	8722	32	180	yes
0+520.00	R	14.67	1.63	5540.9	1.564	9926	2935	9926	32	160	yes
0+780.00	R	4.81	0.53	233.6	0.468	1355	961	1355	32	450	yes
0+800.00	R	5.99	0.67	429.1	0.599	1951	1197	1951	32	450	yes

f'c 28 MPa, fy 420 MPa, phi 0.9, load factor 1.5 on the active thrust (phi' 30 deg, gamma 19 kN/m3, surcharge 10 kPa), min steel ratio 0.0018 - engineer inputs (DEMO VALUE - engineer to replace). Shear, crack width, footing design and detailing remain with the structural engineer.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Retaining wall stem reinforcement

Sheet 25 of 32

Scale: as shown (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

175bc80d01efaeeb29d3307375fe64e2

a486d4e6670d7de3a340c8953107d8d8

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

PRELIMINARY

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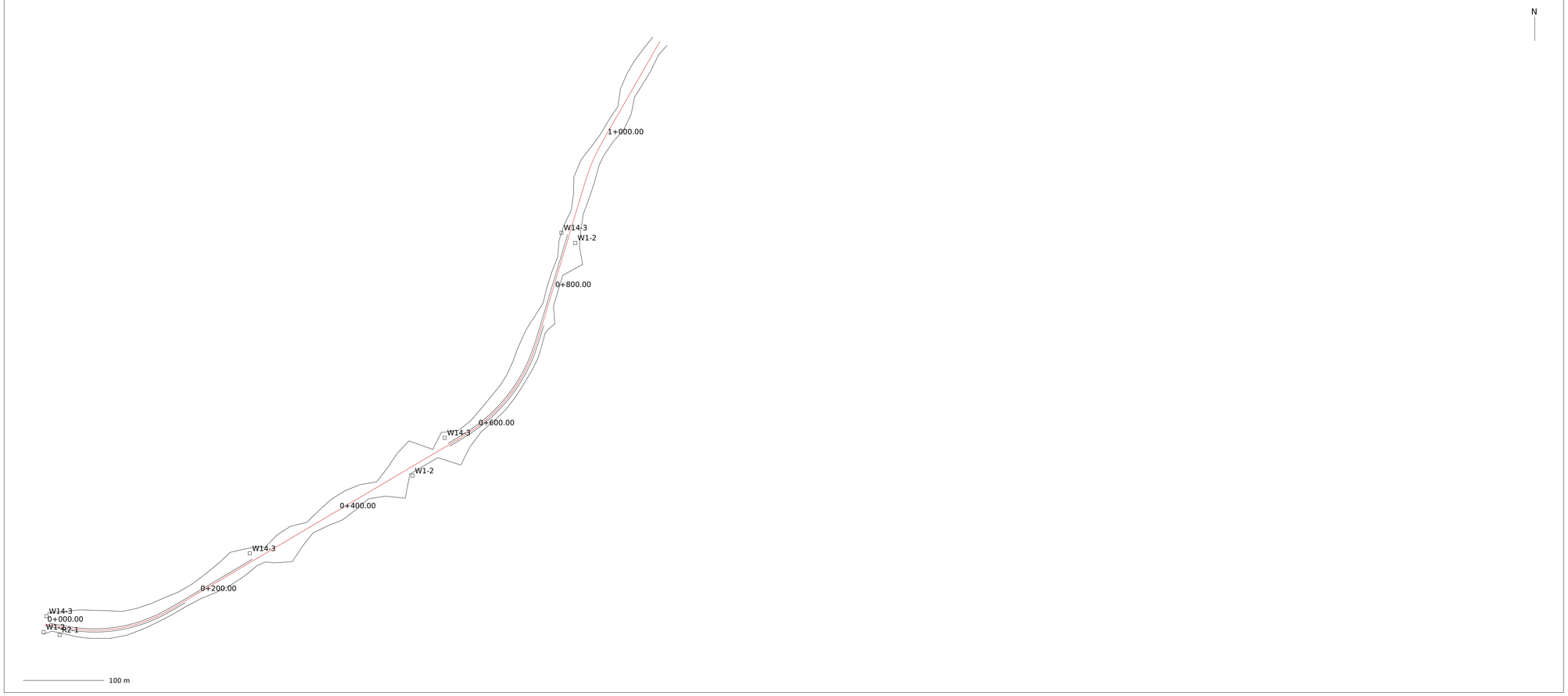
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Signing and pavement marking (concept)



Code	Sign	Station	Side	Reason
W1-2	Curve warning (with advisory speed if ~	0+900.00	right	curve 1 R 180 m < 300 m (engineer threshold)
W1-2	Curve warning (with advisory speed if ~	0+502.02	right	curve 2 R 200 m < 300 m (engineer threshold)
W1-2	Curve warning (with advisory speed if ~	0+862.83	right	curve 3 R 200 m < 300 m (engineer threshold)
R2-1	Speed limit	0+020.00	right	start of project, 50 km/h design speed (posted limit set by the road aut~
W14-3	No passing zone pennant	0+000.00	left	start of no-passing zone (up-station)
W14-3	No passing zone pennant	0+560.00	left	start of no-passing zone (up-station)
W14-3	No passing zone pennant	0+870.00	left	start of no-passing zone (down-station)
W14-3	No passing zone pennant	0+280.00	left	start of no-passing zone (down-station)

No-passing from	to	Direction
0+000.00	0+180.00	up-station
0+560.00	0+755.00	up-station
0+560.00	0+870.00	down-station
0+005.00	0+280.00	down-station

Manual: FHWA MUTCD 11th Edition (2023). PSD 168 m (MUTCD Table 3B-1 (2009 ed.; verify numbering in the 11th ed.), ft→m). Centre line yellow, edge line white.
Manual: FHWA MUTCD 11th Edition (2023)
No-passing zones in both directions from vertical sight distance on the design profile (eye = object height, per the manual) and, where lateral_clear_m is given, from the horizontal sight-line offset M = R(1 - cos(PSD/2R)) on curves. Intersection signing is added by the engineer.
Sign codes are the usual designations in each manual and must be confirmed against the current edition.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Signing and pavement marking (concept)
Sheet 26 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

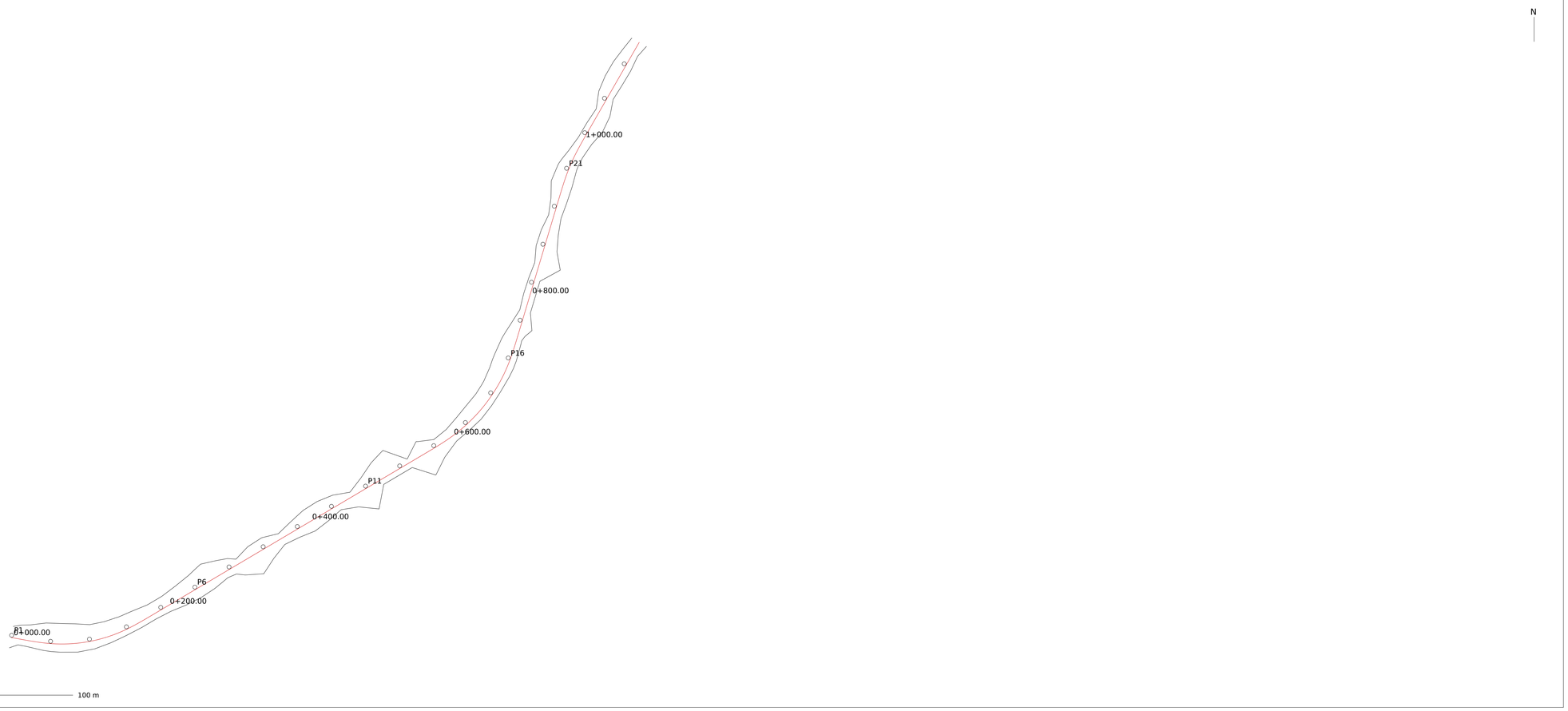
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Lighting layout (concept)



Pole	Station	E	N	Side
P1	0+000.00	573520.49	4139662.71	left
P2	0+048.00	573567.63	4139655.43	left
P3	0+096.00	573614.69	4139658.03	left
P4	0+144.00	573659.40	4139672.94	left
P5	0+192.00	573700.96	4139696.47	left
P6	0+240.00	573742.27	4139720.91	left
P7	0+288.00	573783.58	4139745.36	left
P8	0+336.00	573824.89	4139769.81	left
P9	0+384.00	573866.19	4139794.25	left
P10	0+432.00	573907.50	4139818.70	left
P11	0+480.00	573948.81	4139843.15	left
P12	0+528.00	573990.12	4139867.60	left
P13	0+576.00	574031.37	4139892.08	left
P14	0+624.00	574069.59	4139920.05	left
P15	0+672.00	574100.25	4139955.98	left
P16	0+720.00	574121.51	4139998.16	left
P17	0+768.00	574135.84	4140043.79	left
P18	0+816.00	574149.64	4140089.77	left
P19	0+864.00	574163.43	4140135.74	left
P20	0+912.00	574177.22	4140181.72	left
…	24 poles in ligh…			

S = Φ·CU·LLF/(E·W) = 20000·0.45·0.8/(9·3.50) = 228.6 m; limit 4·H = 48.0 m; used 48.0 m
Concept layout by the lumen method. Photometric design (luminance, uniformity, glare) to IES RP-8 / TAC Guide for the Design of Roadway Lighting / BS 5489-1 + BS EN 13201 / AS/NZS 1158 is by the lighting designer.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Road lighting layout (concept)
Sheet 27 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

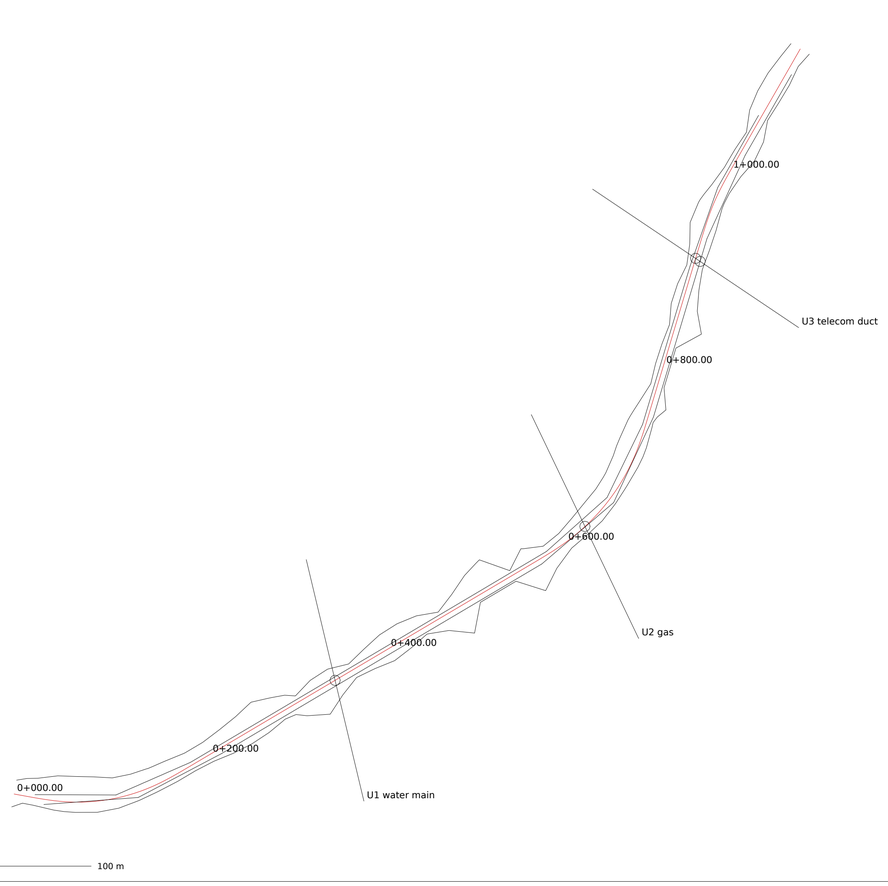
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Utilities and conflicts



Utility	Kind	Owner	Station	Issue	Value m	Action	QL
U1	water main	DEMO UTILITY OWN~	0+340.00	fill increases cover beyond utility limit	5.92	owner to confirm loading / provide access	QL-D (synthetic d~
U2	gas	DEMO UTILITY OWN~	0+420.33	cut/subgrade within clearance of utility	-1.83	pothole (QL-A), then lower/relocate or protect	QL-D (synthetic d~
U3	telecom duct	DEMO UTILITY OWN~	0+790.00	crossing - no vertical conflict found	2.30	protect in place; confirm location	QL-D (synthetic d~
U3	telecom duct	DEMO UTILITY OWN~	crosses Storm pipe P10 with clearance below min 0.3 m		-0.10	adjust pipe grade or relocate utility	QL-D (synthetic d~

Utility records are owner / engineer supplied at the stated quality level (DEMO VALUE - engineer to replace). Clearance 0.3 m and maximum added cover 3 m are engineer inputs. Conflicts at QL-C/D need potholing (QL-A) before design is finalised.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Utility plan and conflict register

Sheet 28 of 32

Scale: as shown (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

PRELIMINARY

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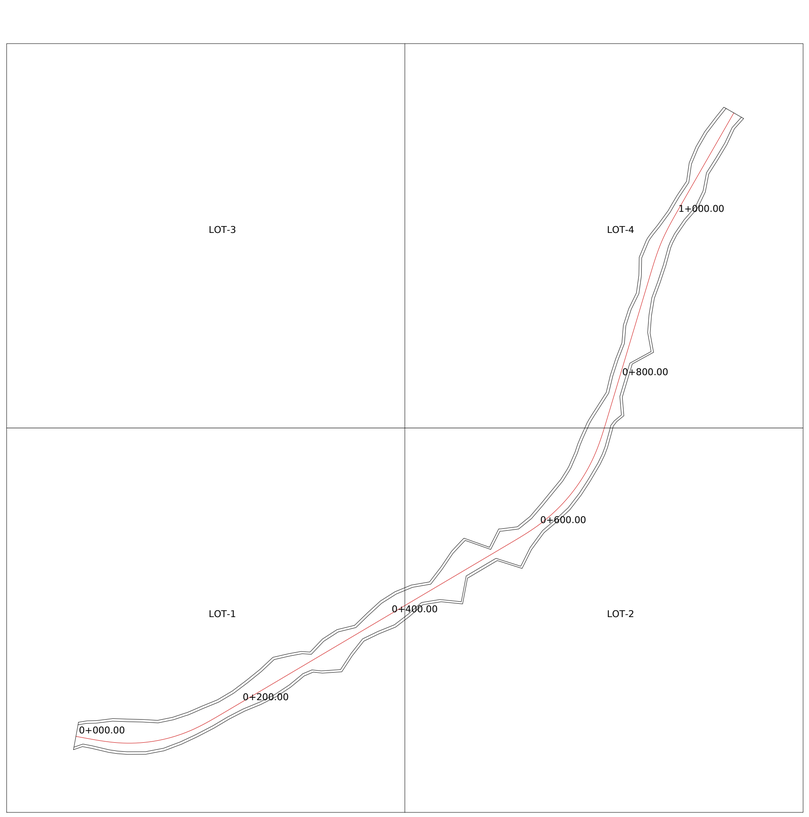
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Land requirement plan



Parcel	Owner	Parcel m2	Take m2	Take %
LOT-1	DEMO PARCEL	204191	13849	6.8
LOT-2	DEMO PARCEL	204191	10213	5.0
LOT-3	DEMO PARCEL	204191	0	0.0
LOT-4	DEMO PARCEL	204191	10755	5.3

Land requirement = daylight lines + 3 m working / maintenance buffer (engineer). Area 34815 m2. Parcel boundaries from surveyor / cadastre data (DEMO VALUE - engineer to replace); legal descriptions and title by the land surveyor.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Right-of-way / land requirement plan
Sheet 29 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

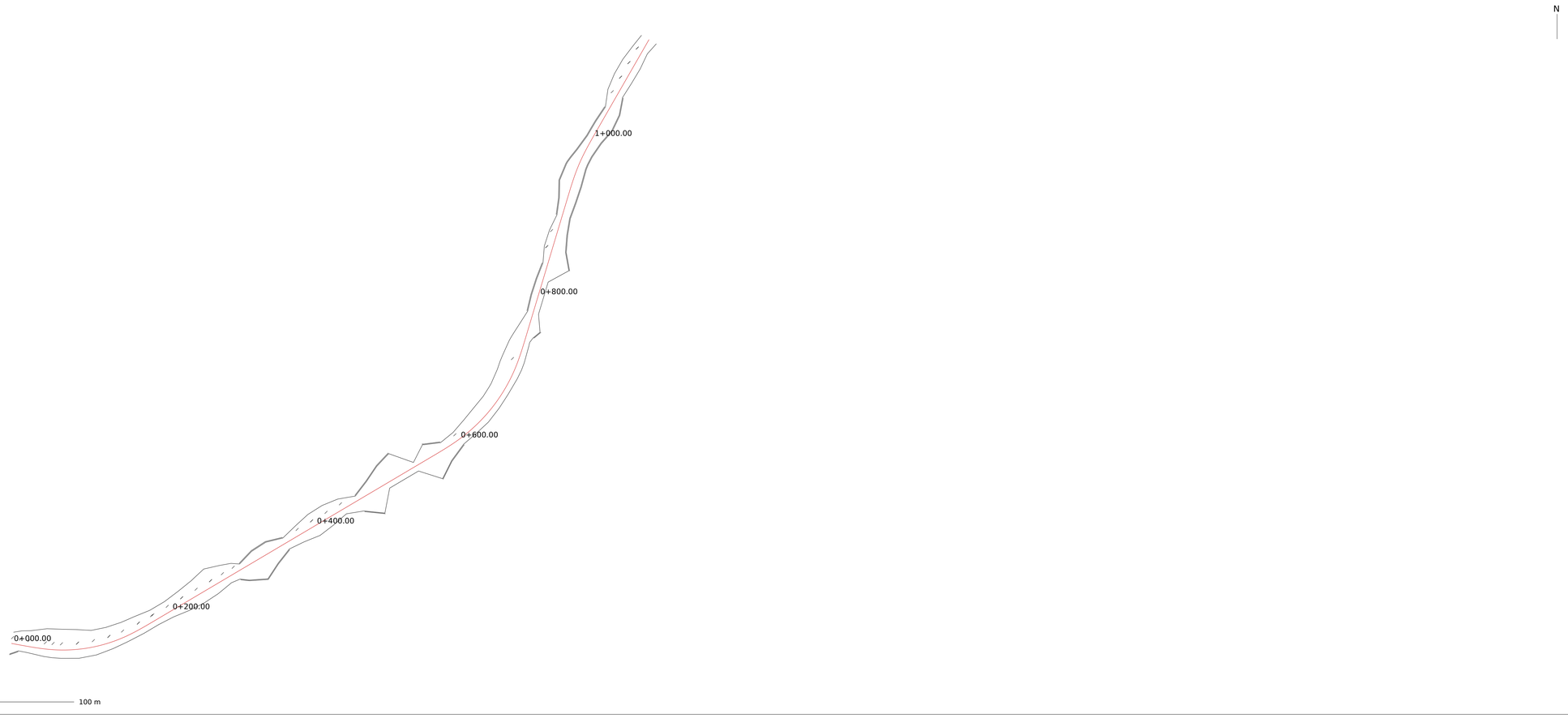
NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE

Erosion and sediment control (concept)



Item	Location	Quantity	Basis
Sediment fence	L 0+300.00 - 0+360.00	62 m	fill toe + offset
Sediment fence	L 0+460.00 - 0+520.00	64 m	fill toe + offset
Sediment fence	L 0+560.00 - 0+580.00	22 m	fill toe + offset
Sediment fence	L 0+780.00 - 0+840.00	60 m	fill toe + offset
Sediment fence	L 0+900.00 - 1+040.00	145 m	fill toe + offset
Sediment fence	R 0+000.00 - 0+009.58	11 m	fill toe + offset
Sediment fence	R 0+291.41 - 0+360.00	77 m	fill toe + offset
Sediment fence	R 0+460.00 - 0+480.00	25 m	fill toe + offset
Sediment fence	R 0+560.00 - 0+602.02	49 m	fill toe + offset
Sediment fence	R 0+751.01 - 0+760.00	11 m	fill toe + offset
Sediment fence	R 0+840.00 - 1+060.00	222 m	fill toe + offset
Check dams	L ditch 0+000.00 - 0+291.41	24 @ 13 m	L = h/S = 0.6/0.0478
Check dams	L ditch 0+380.00 - 0+440.00	5 @ 13 m	L = h/S = 0.6/0.0450
Check dams	L ditch 0+600.00 - 0+760.00	2 @ 120 m	L = h/S = 0.6/0.0050
Check dams	L ditch 0+860.00 - 0+880.00	3 @ 9 m	L = h/S = 0.6/0.0659
Check dams	L ditch 1+060.00 - 1+135.65	7 @ 11 m	L = h/S = 0.6/0.0554
Culvert inlet / outlet protection	2 culverts	4 each	both ends
Stabilised construction entrance	start and end tie-ins	2 each	site access

Concept ESC plan: sediment fence at fill toes + 1 m, check dams at L = h/S (h = 0.6 m), protection at culvert ends, stabilised entrances. Devices, spacing and inspection regime per the agency ESC manual and permit (DEMO VALUE - engineer to replace); the SWPPP / ESC plan is completed by the qualified ESC designer.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Erosion and sediment control plan (concept)
Sheet 30 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
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Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

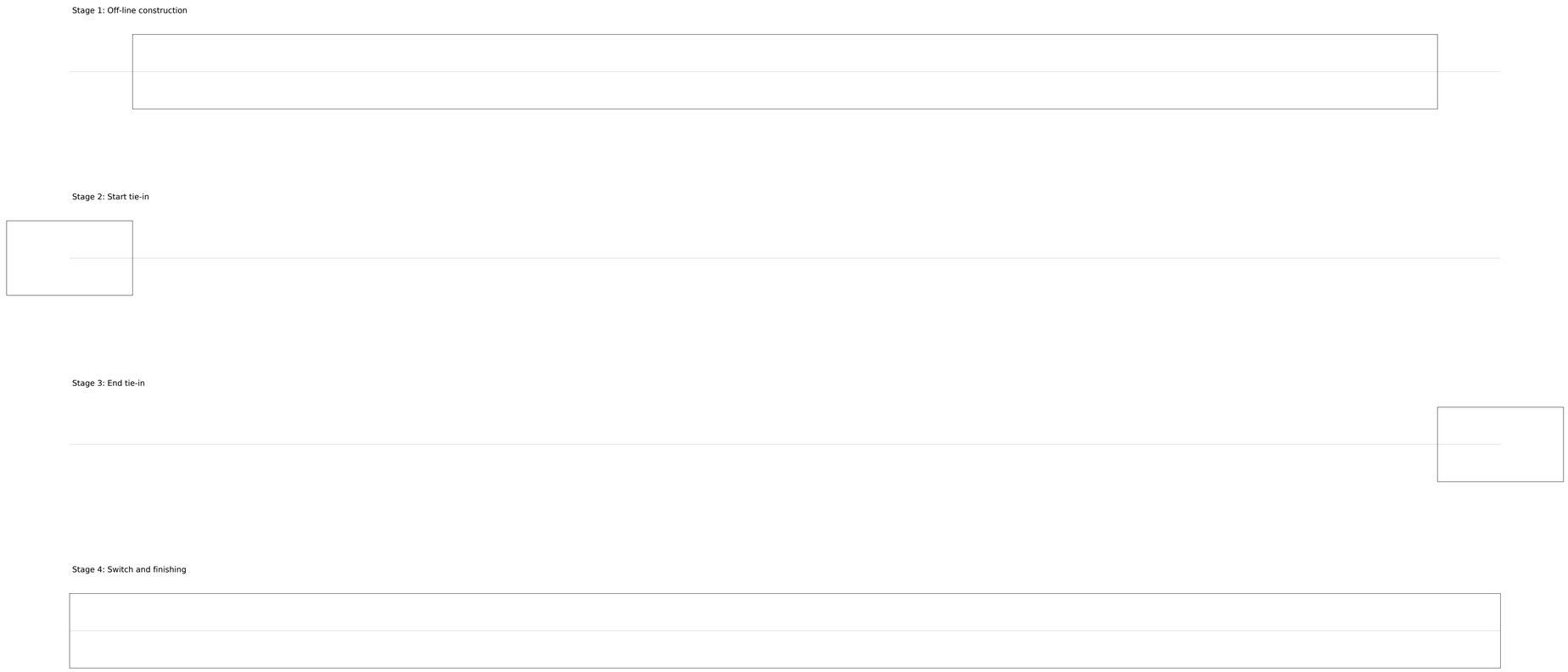
NOT FOR CONSTRUCTION

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Staging diagram (station vs stage)



horizontal: station; bars: work zone extent

Stage	Name	From	To	Traffic	Works	Taper m	Devices
1	Off-line construction	0+050.00	1+085.65	existing network unaffected; site access	earthworks, drainage, culverts, pavemen~		site access signs; construction entrance; fencing
2	Start tie-in	-1+950.00	0+050.00	single-lane alternating flow on the exis~	tie-in pavement, kerb, drainage connect~	167	advance warning series; flagger or portable signals; channelising devices on th~
3	End tie-in	1+085.65	1+185.65	single-lane alternating flow on the exis~	tie-in at the end	161	advance warning series; flagger or portable signals; channelising devices
4	Switch and finishing	0+000.00	1+135.65	traffic on the new alignment; shoulders ~	line-marking, signs, barrier, lighting, ~	54	shoulder taper (L/3); works vehicle with arrow board

MUTCD §6C.08 / Table 6C-4 (2009 ed.; verify numbering in the 11th ed.) taper L = WS²/60 (≤ 40 mph) or WS (≥ 45 mph)
Concept staging only. Detailed TTM / TMP layouts, device spacing and approvals per the agency manual by a qualified TTM designer.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Temporary traffic management staging (concept)
Sheet 31 of 32
Scale: as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 244/244 pass
Audit log head:
175bc80d01efaeeb29d3307375fe64e2
a486d4e6670d7de3a340c8953107d8d8
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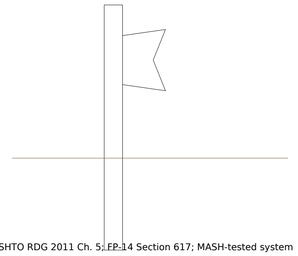
D1 Curb and gutter



Reference: State DOT standard plan; FP-14 Section 609

Schematic only - not to scale. Use the referenced standard drawing.

D2 W-beam guardrail



Reference: AASHTO RDG 2011 Ch. 5; FP-14 Section 617; MASH-tested system

Schematic only - not to scale. Use the referenced standard drawing.

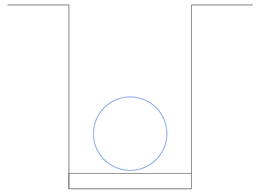
D3 Riprap / ditch lining



Reference: HEC-15 (2005); FP-14 Sections 251, 629

Schematic only - not to scale. Use the referenced standard drawing.

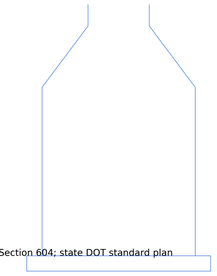
D4 Pipe trench and bedding



Reference: FP-14 Section 602; pipe manufacturer / AASHTO LRFD Sec. 12

Schematic only - not to scale. Use the referenced standard drawing.

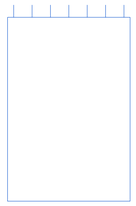
D5 Manhole



Reference: FP-14 Section 604; state DOT standard plan

Schematic only - not to scale. Use the referenced standard drawing.

D6 Grate inlet



Reference: HEC-22 (2009) inlet capacity; FP-14 Section 604

Schematic only - not to scale. Use the referenced standard drawing.

D7 Pavement edge / shoulder



Reference: AASHTO 93 design; FP-14 Sections 301, 401

Schematic only - not to scale. Use the referenced standard drawing.

D8 Sidewalk



Reference: FP-14 Section 615; PROWAG accessibility

Schematic only - not to scale. Use the referenced standard drawing.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Standard details atlas (references)

Sheet 32 of 32

Scale: NTS (A1)

Standards: Us

Design speed: 50 km/h

Checks: 244/244 pass

Audit log head:

175bc80d01efaeeb29d3307375fe64e2a486d4e6670d7de3a340c8953107d8d8

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

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